Assignment 3 Report

Course: 203.3630 — Artificial Intelligence Lab, University of Haifa
Lecturer: Shay Bushinsky
Assignment: Experiment 3 — Multi-Step Heuristics, Meta-Heuristics and Swarm Intelligence (CVRP), with generative evolutionary AI (Rush Hour GP/GEP)
Submitted by: Mohamed Awad — ID 207731647, and Salah Hamam — ID 318827417
Submission: pair submission
Date: July 2026
Repository: https://github.com/mohammedawad99/ai-lab-assignment3

All numbers in this report come from the generated result files under results/final_experiments/ (final run with seeds 42, 43, 44 from configs/final_experiment_plan.json). Nothing was filled in by hand without a matching CSV row; scripts/extract_report_numbers.py prints the same facts for checking.

1. Introduction

The assignment has two parts. Part A asks for six search algorithms (Simulated Annealing, Tabu Search, Ant Colony Optimization, a Genetic Algorithm with an Island Model, Adaptive Large Neighborhood Search, and Branch & Bound / Limited Discrepancy Search), first validated on the continuous Ackley function as a warm-up and then compared on six official CVRP benchmark instances, together with an explicit multi-stage heuristic. Part B asks for generative AI: evolving Rush Hour heuristic functions with both Genetic Programming (GP) and Gene Expression Programming (GEP), where every candidate heuristic is judged by actually running A* with it.

All stochastic runs use the fixed seeds 42, 43 and 44 from the final experiment plan, so every number below is reproducible. This 3-seed final run is the canonical source of every headline table in this report; the only 8-seed numbers are in the separate robustness study (Section 4.3, seeds 42–49), which is labeled as such and is not the source of any headline value.

The final version of the project contains four layers on top of the first complete run: a controlled tuning pass (validated on all six instances before acceptance) that improved SA, GA-Island and ALNS; a B&B/LDS small-instance mode; a local-search performance pass (O(1) delta 2-opt) followed by advanced local-search moves (inter-route swap, Or-opt, 2-opt*, candidate-list neighborhoods) gated to the larger instances; and, on Part B, a measured 14-puzzle hard Rush Hour benchmark plus an 8-seed CVRP robustness analysis and the optional no-A* bonus: GP/GEP evolved as direct planners (Section 8.2). The final rerun with all accepted settings improved the best CVRP gaps to 2.95% on A-n80-k10, 23.01% on X-n101-k25 and 5.51% on M-n200-k17 with no project-best regressions, and the Ackley results were left unchanged. All numbers in this report come from that final rerun.

On top of the six comparison algorithms, the project implements the assignment's iterated-local-search requirement as an explicit ILS driver (Section 6.7): a separate solver that alternates a ruin-and-recreate perturbation (exploration) with the project's strongest deterministic local-search stack (exploitation) and applies an acceptance rule between complete local optima. ILS was benchmarked under exactly the same seeds, budgets and per-instance timeouts as the six algorithms and is reported separately from the six-algorithm comparison tables; it produced the project's overall best results on two instances (1.5219% on A-n80-k10 and 4.9072% on M-n200-k17, Section 6.7).

2. Implementation Overview

2.1 Project structure

The code is a plain Python package: src/common/ (timing), src/ackley/ (function + the six adaptations), src/cvrp/ (model, parser, cost, validation, baseline) with src/cvrp/solvers/ (the six algorithms), src/rushhour/ (board, A*, safe evaluator, GP/GEP comparison), src/gp/ and src/gep/ (separate frameworks), and src/experiments/ (runners, summaries, report assets). scripts/ holds the CLI entry points and tests/ the pytest suite (390 tests at the time of this report).

2.2 Reproducibility and command-line interface

Every run takes its inputs from CLI arguments — instance paths, seeds, budgets and timeouts are never hardcoded. Each experiment row is written to CSV with its seed, budget, timeout, costs, feasibility, errors and timing. The final settings live in configs/final_experiment_plan.json and the runner scripts/run_final_experiments.py is resumable (finished raw CSVs are skipped on a rerun).

2.3 Validation and safety checks

Every CVRP solution is validated: routes start and end at the depot, no customer is missing or duplicated, capacity is respected, and the number of used routes must not exceed the vehicle count. Rows report feasible and the exact validation errors — nothing is silently fixed. Rush Hour heuristic evaluation runs A* under a node cap, a per-puzzle time cap and a total time budget, and an exception inside a candidate heuristic is recorded as a failed puzzle instead of crashing the run.

2.4 Solution output format and the 80.64 sanity check

Every CVRP solver prints the assignment's output format: the heuristic value (total cost) on the first line, then one line per used vehicle with the visited city indices in order, starting and ending at the depot index 0 (--include-unused-vehicles also prints 0 0 lines for idle vehicles).

On the assignment's 5-city example (depot at the origin, four unit-demand-3 cities, capacity 10) the multi-stage baseline reproduces the expected sub-optimal reference solution and cost:

```
80.64
0 1 2 3 0
0 4 0
```

This sanity check is automated: python a3.py sanity runs it, and the I/O tests assert the exact first line. Elapsed and CPU times are printed with every run (Section 9 reports both for all algorithms).

3. Part A — Ackley Function

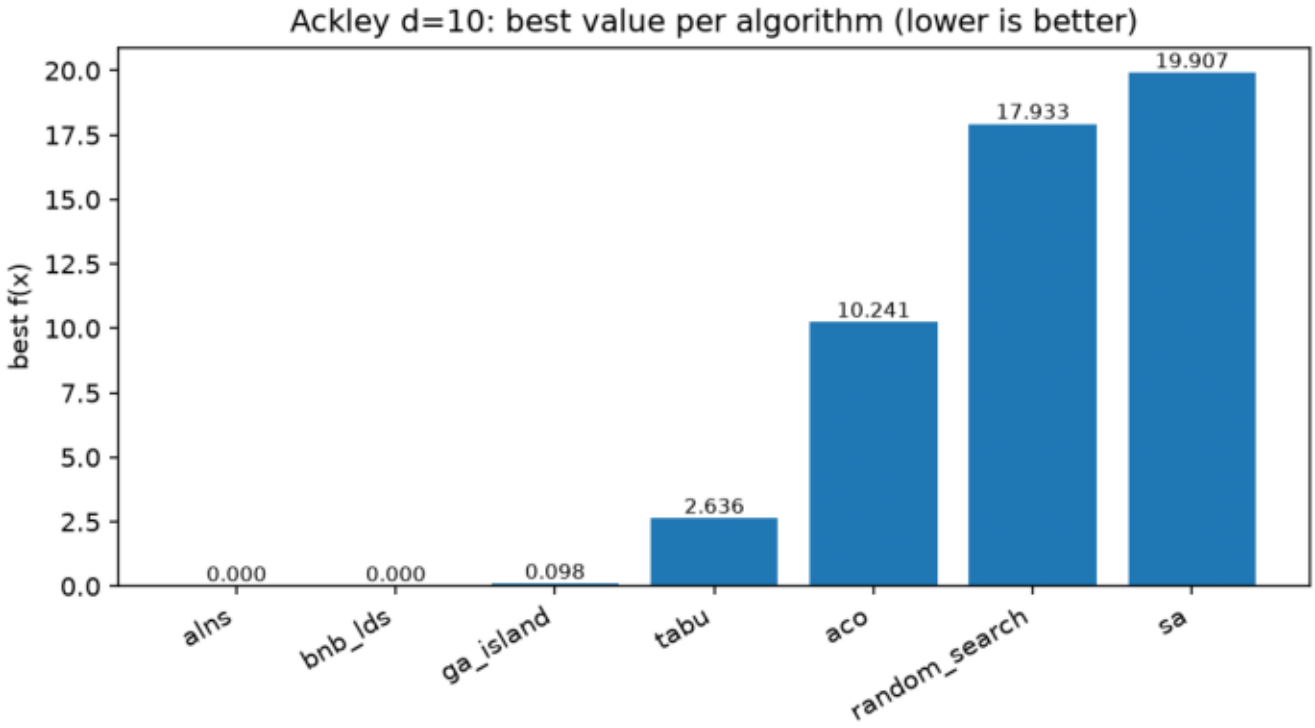
The Ackley function for d dimensions:

```
f(x) = -a * exp(-b * sqrt((1/d) * sum(x_i^2)))
       - exp((1/d) * sum(cos(c * x_i))) + a + e
```

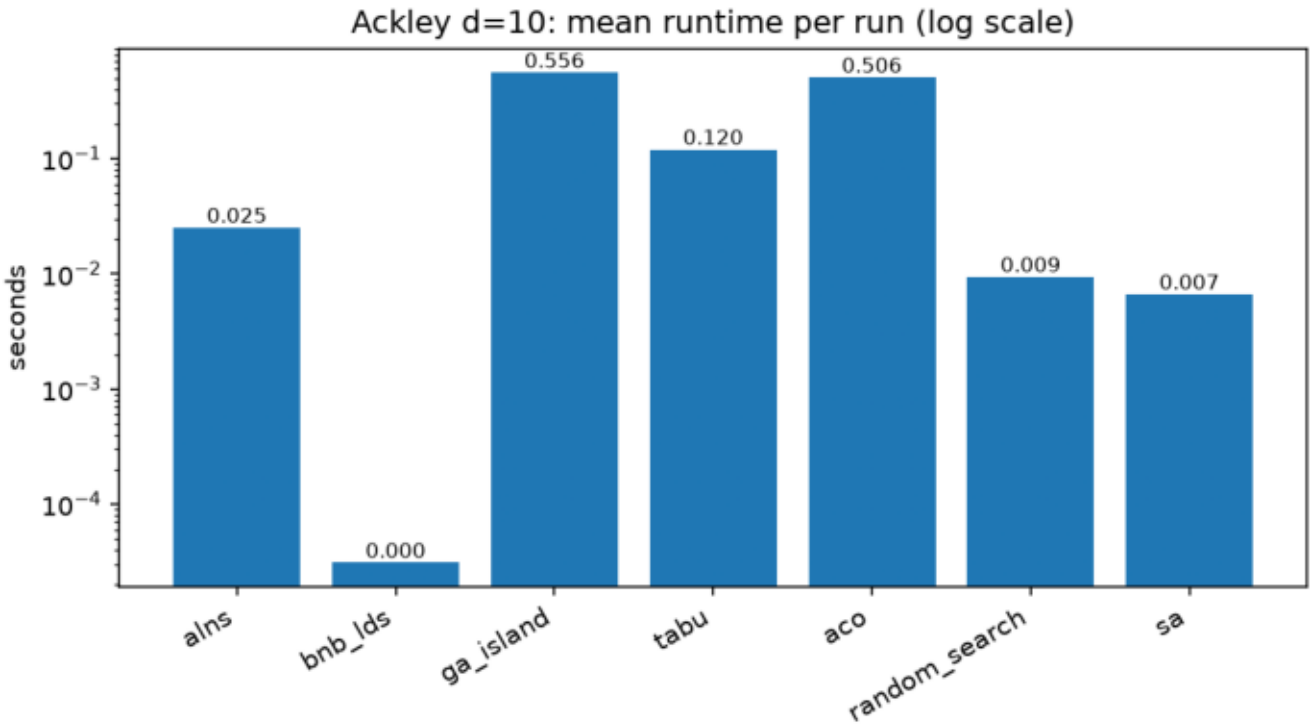
with a = 20, b = 0.2, c = 2*pi, dimension d = 10, bounds x_i in [-32.768, 32.768], and the known optimum f(0, ..., 0) = 0.

Final results (3 seeds per algorithm, budget 500, timeout 60 s), from summary/ackley_d10_summary.csv:

algorithm	runs	best_value	mean_best_value	std_best_value	mean_dist_from_origin	mean_elapsed (s)
ackley_alns	3	0.000000	0.000000	0.000000	0.000000	0.025
ackley_bnb_lds	3	0.000000	0.000000	0.000000	0.000000	0.000
ackley_ga_island	3	0.098	0.100	0.002	0.063	0.556
ackley_tabu	3	2.636	3.018	0.359	1.466	0.120
ackley_aco	3	10.241	11.763	1.406	11.187	0.506
random_search	3	17.933	18.320	0.481	30.278	0.009
ackley_sa	3	19.907	20.041	0.118	46.706	0.007



The figure shows the best value each algorithm reached over its three seeds, sorted from best to worst. The spread is huge — from exactly 0 to almost 20 — which mostly reflects how well each adaptation fits a continuous function, not a universal ranking of the methods.



Runtime (log scale) shows the other side: SA and random search are almost free, while ACO and GA-Island pay for population/colony bookkeeping. Note that B&B/LDS looks instant only because its greedy zero-first bin order finds the origin immediately and the search stops early.

Short analysis. ALNS and B&B/LDS reached 0.000000 (six decimals) on every seed — but both benefit from knowing the optimum is at the origin: the ALNS "toward zero" repair operator and the B&B/LDS bin ranking by distance to zero point straight at it, so this says more about the adaptation than about general optimization strength. GA-Island got close (?0.1) without such a hint. Tabu Search reached ?2.6. ACO's coarse bins (10 per dimension) limit its precision. SA with the untuned parameters (initial temperature 10, step scale 1) actually ended worse than plain random search — an honest negative result: 500 local Gaussian steps from a random corner of a 10-dimensional box are simply not enough without parameter tuning.

4. Part A — CVRP

Official instances and best-known solutions (BKS):

instance	BKS cost
P-n16-k8	450
E-n22-k4	375
A-n32-k5	784
A-n80-k10	1763
X-n101-k25	27591
M-n200-k17	1275

The BKS values are used only to compute the gap percentage (100 · (cost ? BKS) / BKS); the algorithms never see them.

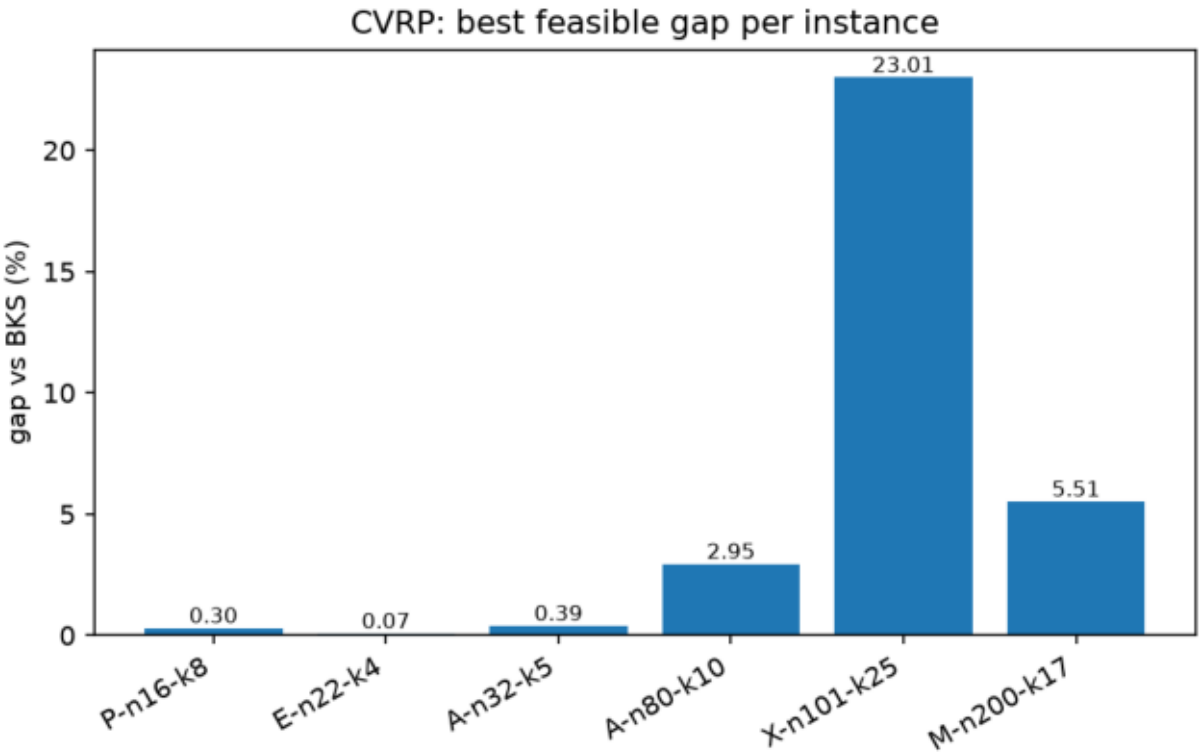
The final run uses tuned settings accepted after a separate validation pass: SA runs 50× more (very cheap) iterations with a slower cooling schedule and a higher start temperature; GA-Island uses population 30 with mutation 0.3; and ALNS gained extra destroy/repair operators (Shaw-style related removal, full-route removal, segment removal, regret-3 insertion). Because the enhanced ALNS regressed on M-n200-k17 during validation, the final rerun executes both ALNS variants (144 raw rows in total: 8 algorithm rows × 6 instances × 3 seeds) and the reported "alns" result

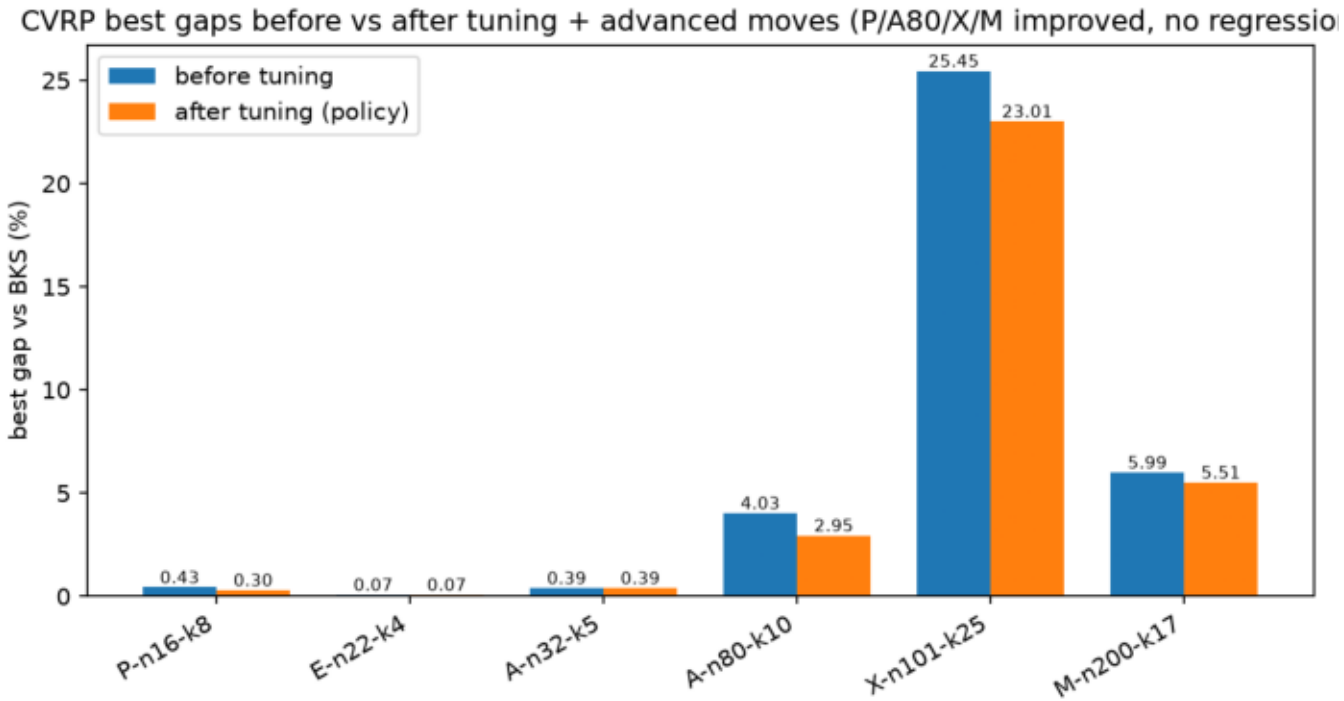
follows a rule declared *before* the rerun: enhanced everywhere except M-n200-k17, which uses the basic variant. Nothing is cherry-picked after the fact and both variants' raw rows are kept.

On top of the tuning, the final run includes a two-step local-search upgrade. First, 2-opt was rewritten with O(1) delta evaluation (only the four changed edges are priced per candidate move instead of recomputing the whole route) — proven byte-identical to the old implementation by tests, so it changed runtime, not results. Second, that headroom paid for advanced moves: inter-route relocate and swap, Or-opt (segment relocate of length 2–3), and 2-opt* (cross-route tail exchange), combined into an intensification pass that ALNS applies to new best solutions and GA-Island applies to its elite every 10 generations (with the polished chromosome reinjected). Neighborhoods can be restricted to precomputed k-nearest candidate lists (k = 10 accepted; k = 10/20/full produced identical solutions in validation). Because validation showed a small regression on the tiny instances (P-n16-k8 0.30?0.39), the advanced pass is gated by instance size — it only activates for instances with at least 60 customers, a customer-count threshold like the B&B small-instance mode, not an instance name list — so the small instances keep their already-validated behavior exactly.

Best feasible result per instance (all 144 rows feasible):

instance	BKS	best algorithm	best cost	best gap	before tuning + advanced
P-n16-k8	450	sa (tuned)	451.34	0.2967%	0.43%
E-n22-k4	375	sa (tuned)	375.28	0.0746%	0.07%
A-n32-k5	784	alns	787.08	0.3931%	0.39%
A-n80-k10	1763	alns (enhanced + advanced)	1814.95	2.9466%	4.03%
X-n101-k25	27591	ga_island (tuned + advanced)	33938.66	23.0063%	25.45%
M-n200-k17	1275	alns (basic + advanced)	1345.30	5.5139%	5.99%

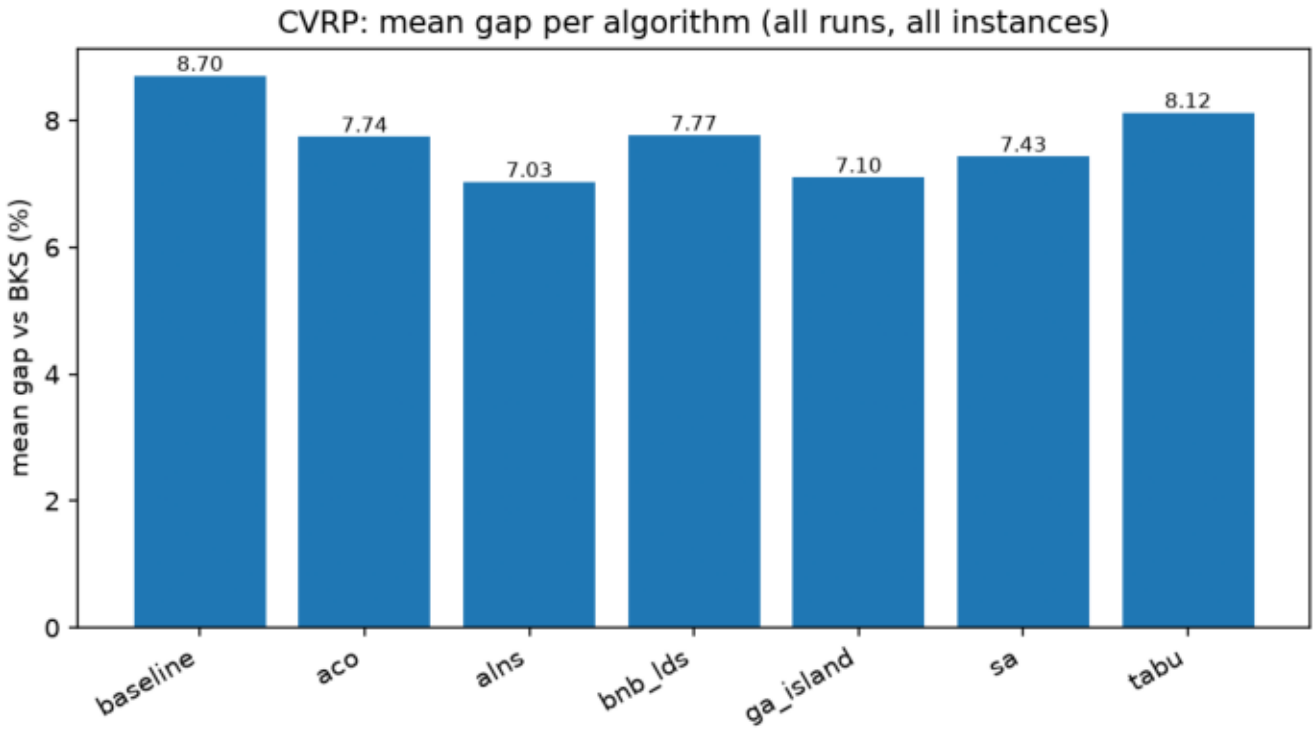




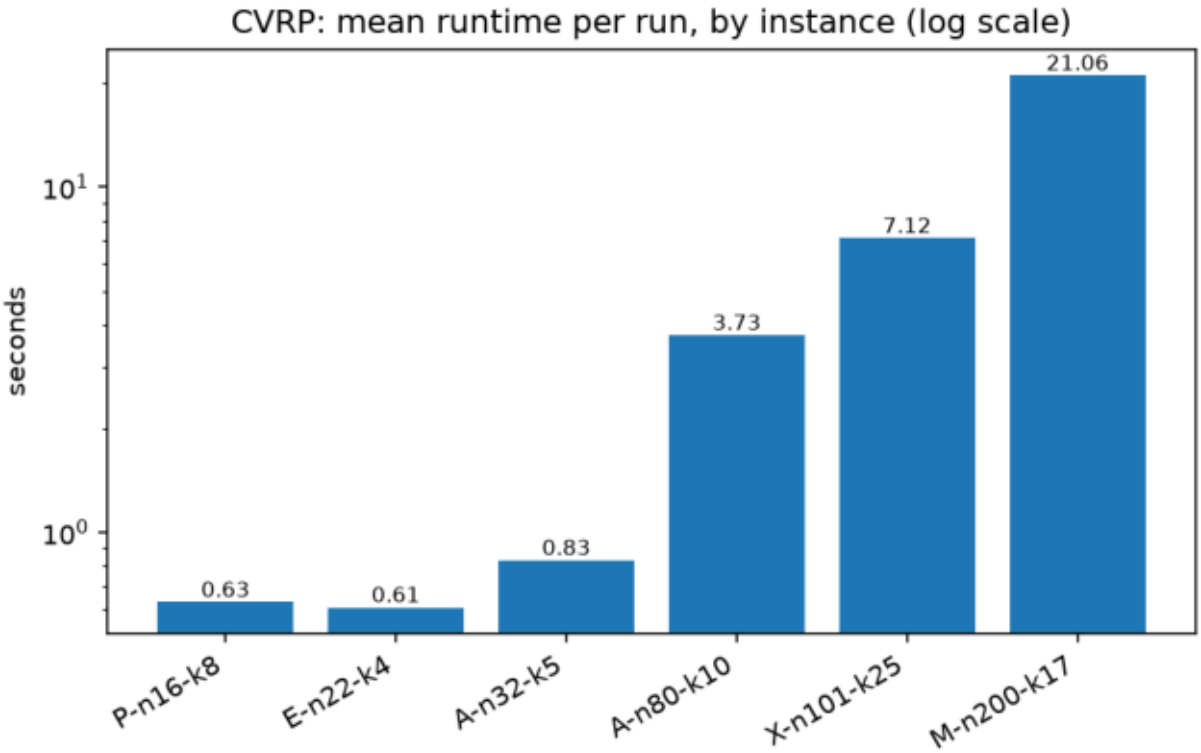
The before/after chart shows the combined effect of tuning plus the advanced moves: P-n16-k8 (0.43?0.30), A-n80-k10 (4.03?2.95), X-n101-k25 (25.45?23.01) and M-n200-k17 (5.99?5.51) improved, E-n22-k4 and A-n32-k5 stayed exactly where they were, and no instance got worse. The X-n101-k25 column still stands out; that is a capacity-packing property of the instance, explained below, not an algorithm bug.

Per algorithm (mean of best-of-seeds gaps over the six instances, policy view):

algorithm	mean best gap	beats baseline on
alns (policy + advanced)	5.851%	6/6 instances
sa (tuned)	6.740%	4/6
ga_island (tuned + advanced)	6.958%	6/6
aco	7.558%	4/6
bnb_lds (small-instance mode)	7.765%	2/6
tabu	8.092%	5/6
baseline	8.703%	—



ALNS keeps the lowest mean best gap (5.85%), and with the advanced pass GA-Island now also beats the multi-stage baseline on all six instances — its memetic polish finally moved it off the baseline on A-n80-k10 (4.66% vs the baseline 4.95%) and it holds the overall best X-n101-k25 result. The tuned SA remains the biggest single tuning mover (8.50% to 6.74% purely by spending its unused time budget on more iterations with a longer schedule). One honest caveat: the advanced pass changed the enhanced ALNS trajectory on X-n101-k25 for the worse (25.01?25.88), which is why the ALNS mean ticked up from 5.80% to 5.85% even though ALNS improved on A-n80-k10 and M-n200-k17 — the regression is disclosed in Section 4.4. B&B/LDS keeps its small-instance mode results: it ties the project-best on P-n16-k8 (0.30%) and E-n22-k4 (0.07%) in about a second each — but on the four larger instances it still returns its starting incumbent, which is stated as-is (overall mean 7.77%, beating the baseline on 2/6).

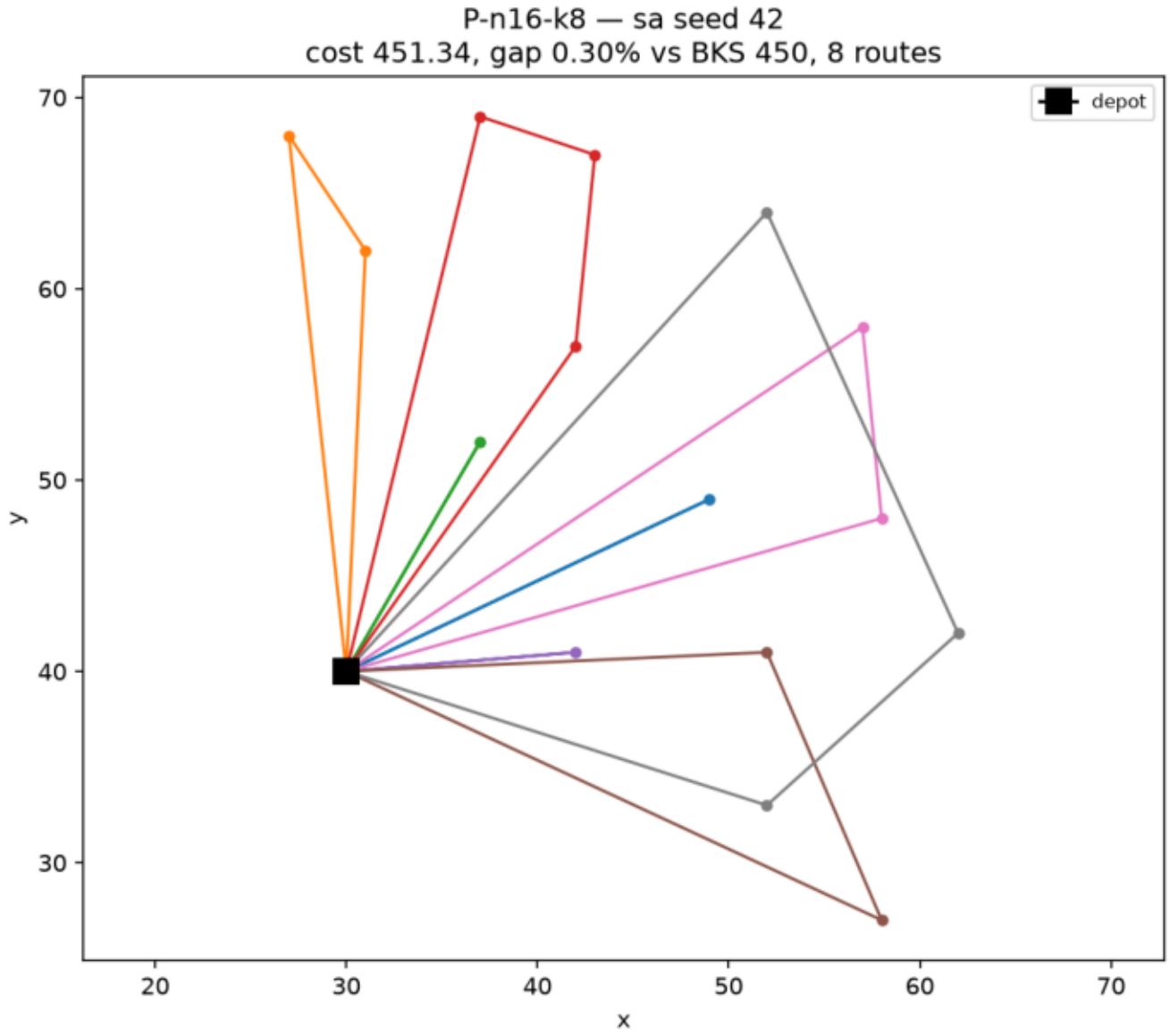


Mean runtime per run grows with instance size as expected (log scale). The values are averages over all seven algorithms, so the cheap methods (SA, baseline) pull the mean down; ACO alone accounts for most of the time on the two largest instances. These are wall-clock (elapsed) means; the matching CPU-time means are reported per algorithm in the Section 9 table — elapsed and CPU time are recorded together for every single run.

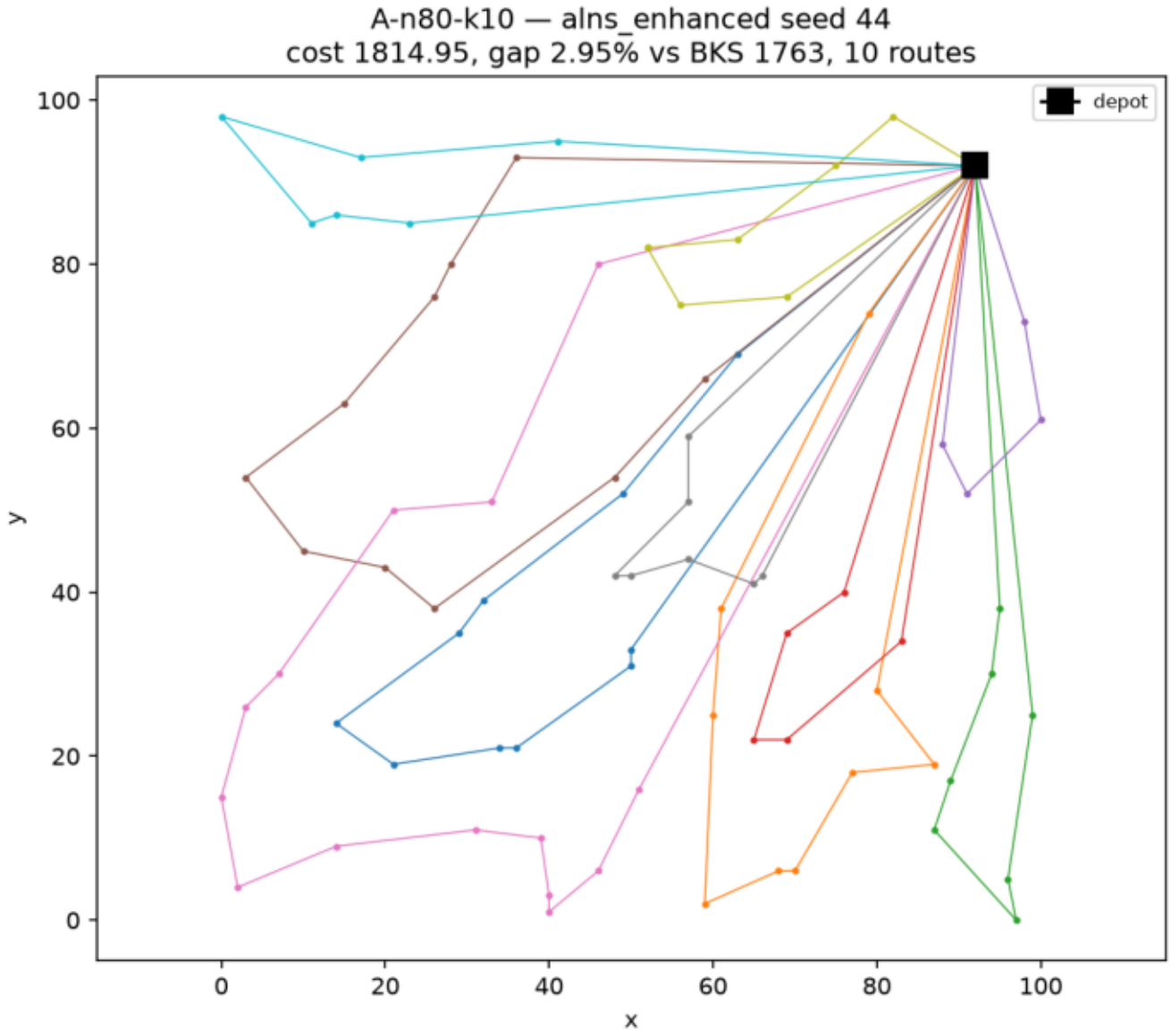
The tables above are the six-algorithm comparison required by the assignment and are unchanged by the later ILS stage. The explicit ILS driver (Section 6.7) ran separately under the same seeds, budgets and timeouts; it improved the project-best gaps on A-n80-k10 (2.9466% ? 1.5219%) and M-n200-k17 (5.5139% ? 4.9072%) while not beating GA-Island on X-n101-k25.

4.1 Route visualizations

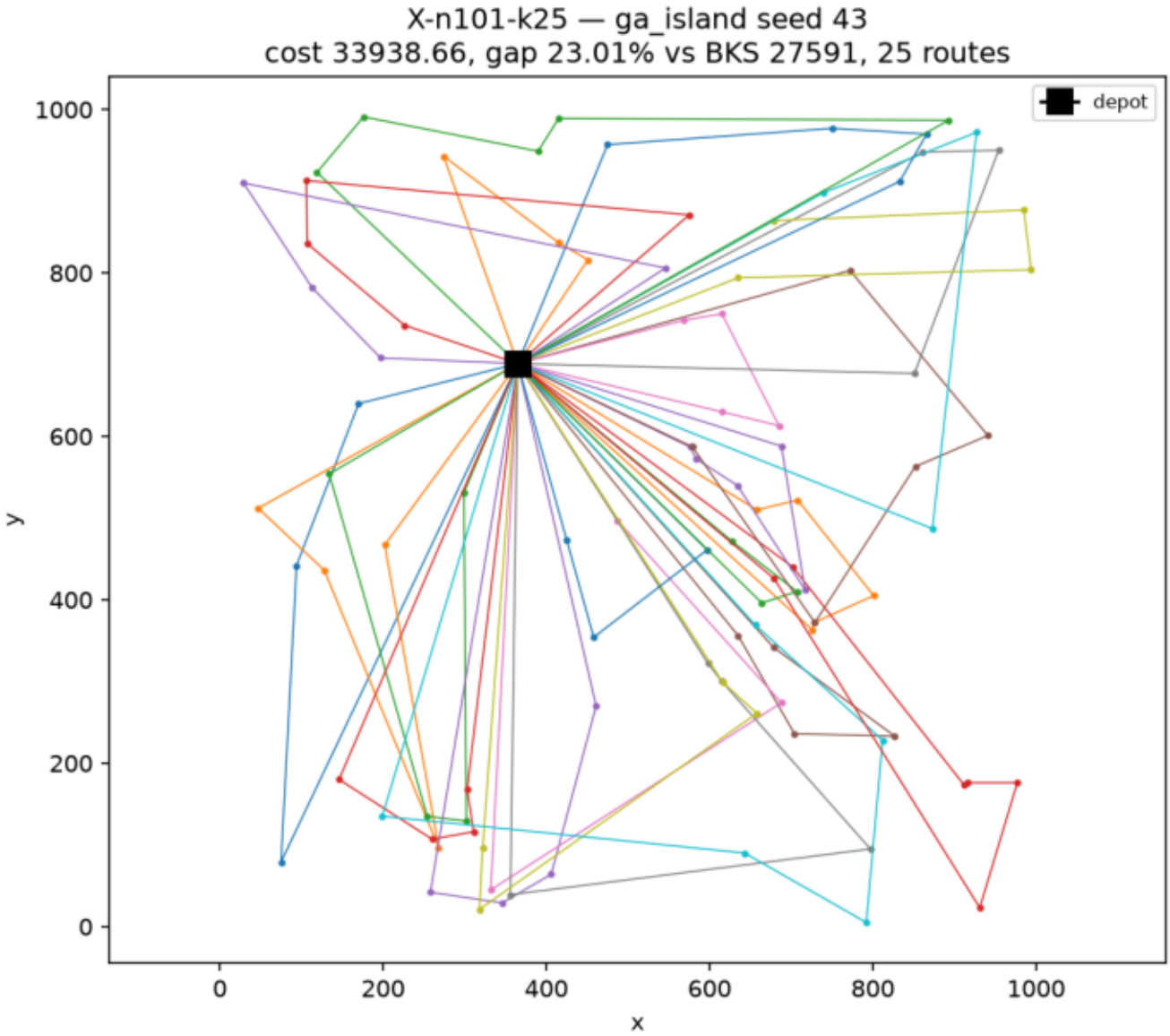
Route plots are included to check feasibility visually on top of the validator checks: every tour must start and end at the black depot square, and the route count in the title must not exceed the fleet size.



P-n16-k8 (tuned SA, seed 42, cost 451.34): eight short routes, each serving only one or two customers — with capacity 35 and demands up to 30-plus, most vehicles can take very little, which is why the instance needs all eight vehicles despite having only 15 customers.

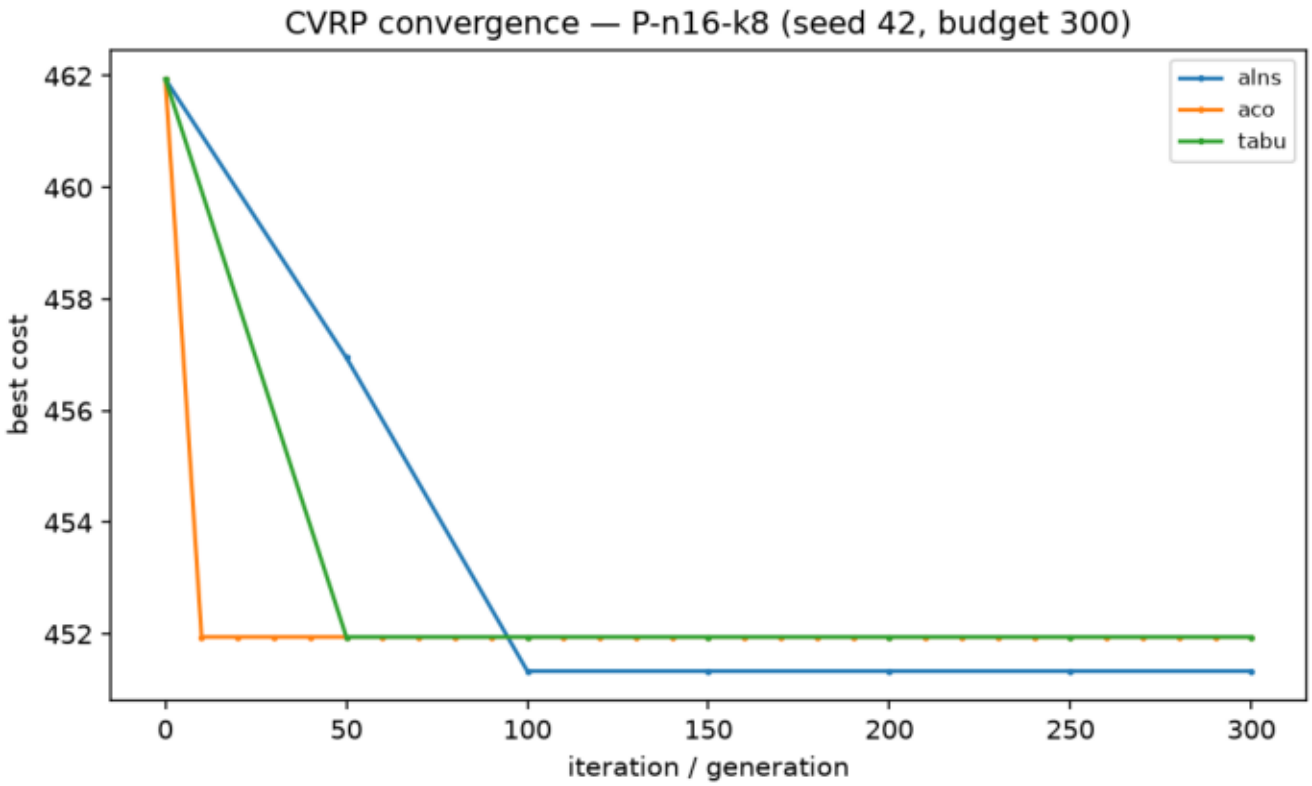


A-n80-k10 (enhanced ALNS with the advanced pass, seed 44, cost 1814.95): ten routes fanning out of the depot in clean geographic sectors. A few crossings between neighboring routes remain — visual evidence of the remaining ~2.9% gap that even the inter-route moves cannot remove within the budget.

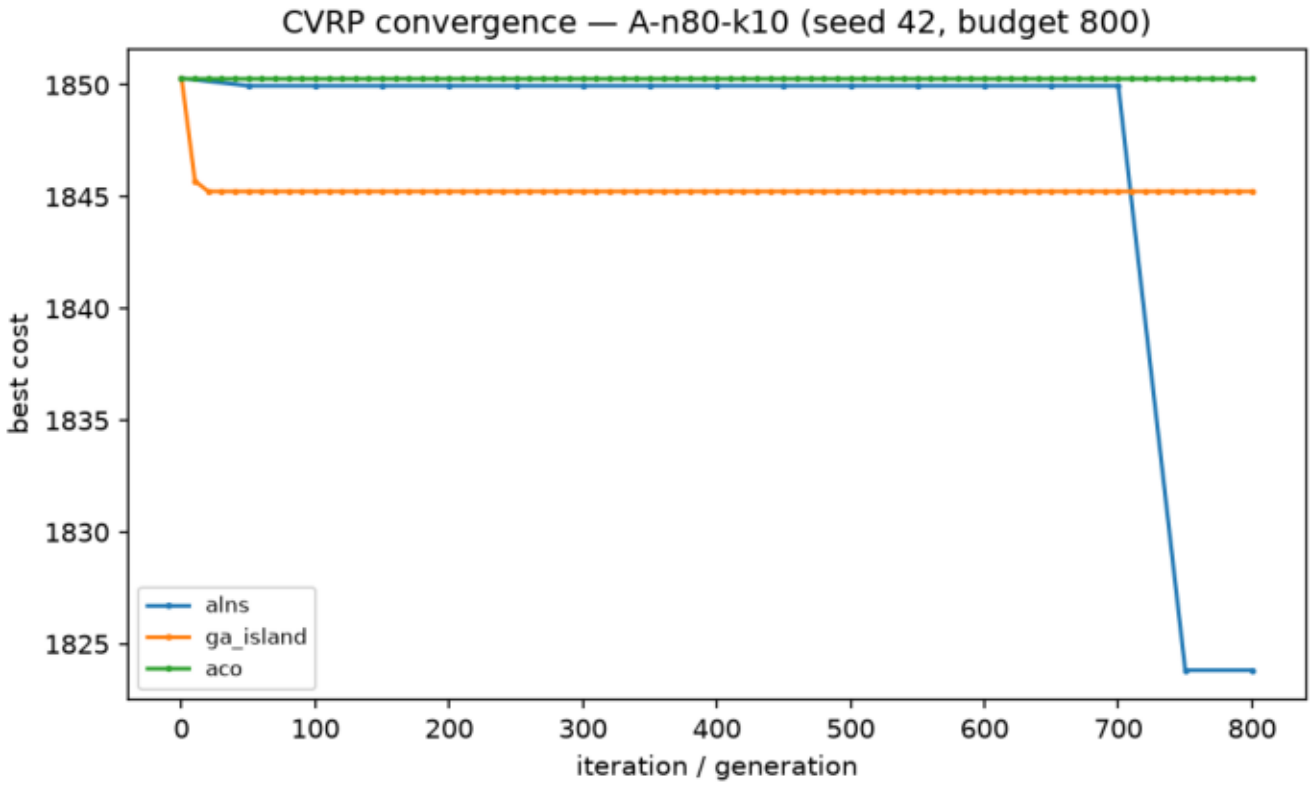


X-n101-k25 (tuned GA-Island with the advanced pass, seed 43, cost 33938.66): the plot is dense
because the instance is large (100 customers) and extremely tight (25 nearly-full routes), so it is harder to inspect visually — long criss-crossing legs are exactly what a load-driven packing looks like when geometry has to take second place to capacity.

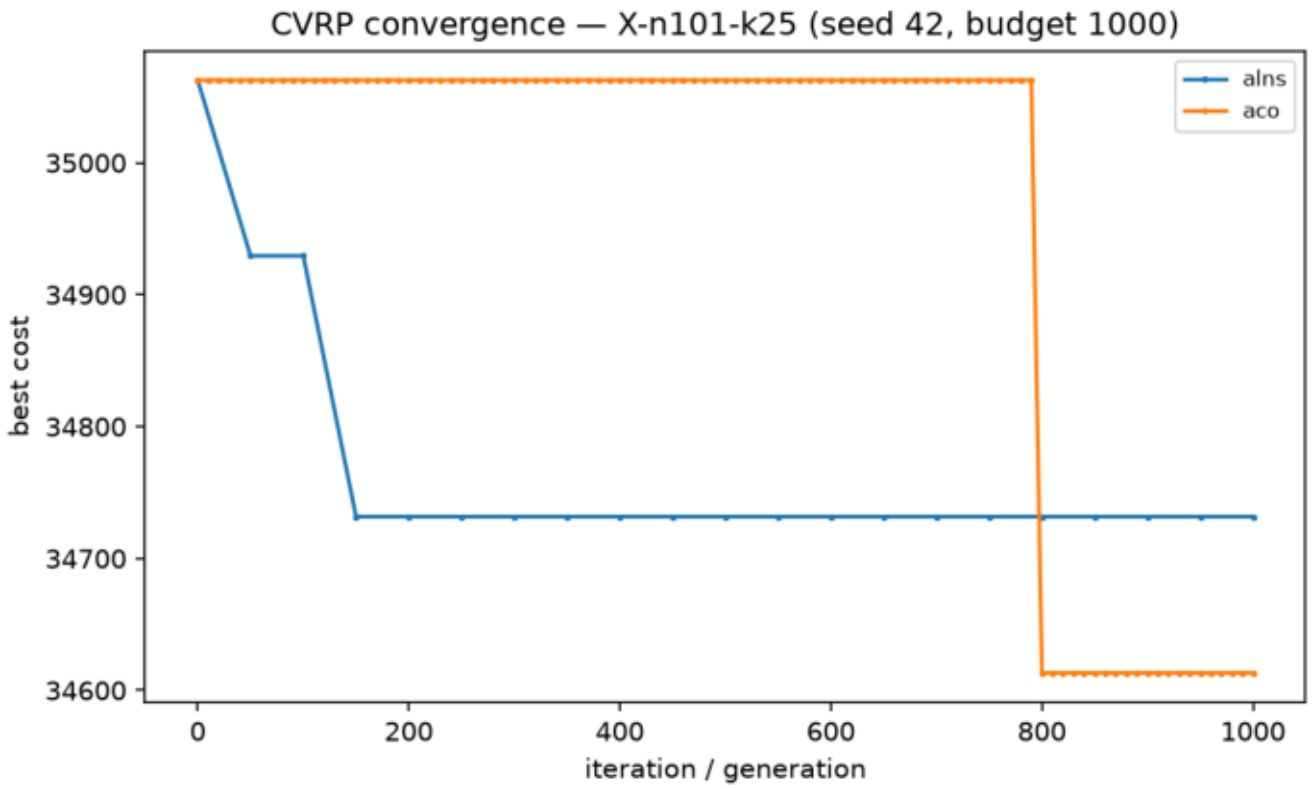
4.2 Convergence behavior



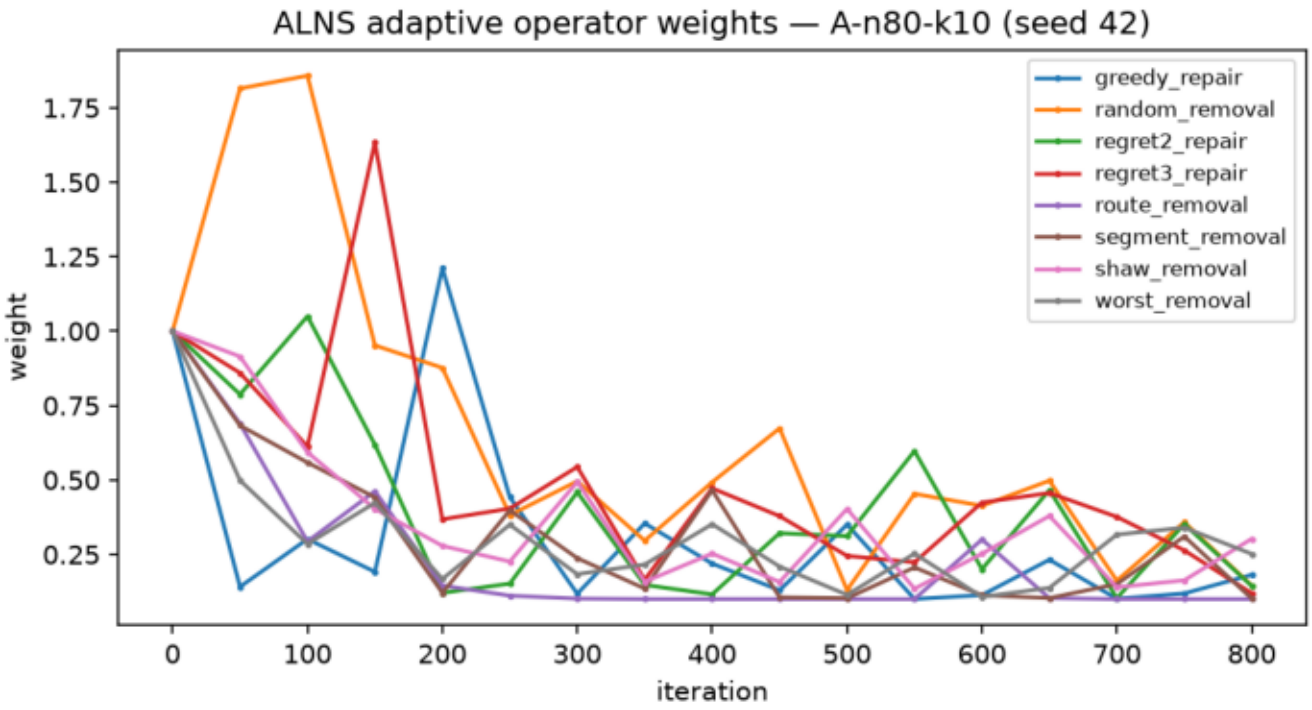
On the small instance every method starts from the repaired baseline (461.94) and improves quickly: ACO and Tabu reach 451.95 within the first 50 iterations, and the enhanced ALNS goes one step further to 451.34 for this seed (the advanced pass is gated off here — 15 customers is far below the 60-customer threshold). After that the curves are flat — the instance is essentially solved as far as these operators can take it.



On A-n80-k10 most of the quality still comes from the multi-stage baseline (1850.30), but with the advanced intensification the enhanced ALNS pulls away to 1823.82 for seed 42 (its best seed reaches 1814.95) and the tuned GA-Island — which used to sit on the baseline for this seed — now steps down to 1845.24 through its memetic polish, while ACO stays on the baseline. The strong start compresses the visible improvement — the y-axis spans a handful of cost units.



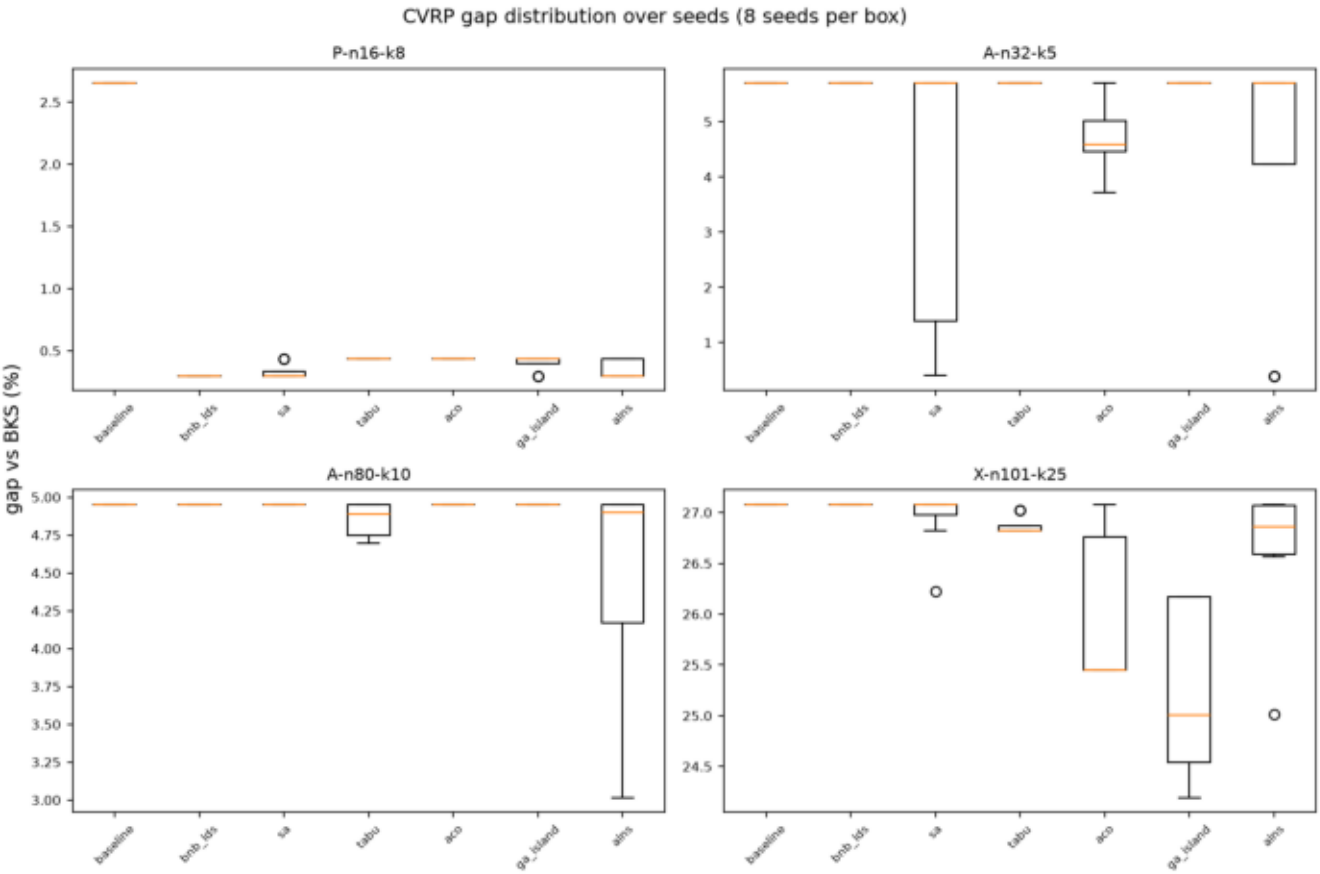
On X-n101-k25 the curves confirm the packing story: the enhanced ALNS route-removal operator can restructure whole routes and improves the subset-sum start to 34731.66 for this seed (basic ALNS could not move at all in the 3-unit-slack packing), and ACO's constructive ants reach 34613.14 — for this particular seed ACO actually ends below ALNS, which is the seed-level face of the ALNS-on-X regression disclosed in Section 4.4. Neither gets anywhere near the BKS — improvement is still limited after feasible packing, as discussed above.



The ALNS adaptive layer is visibly working, now over the full enhanced pool (five destroy and three repair operators): weights move away from their initial 1.0 within the first ~50 iterations and then stabilize once improvements dry up. The drift toward the reject-score floor is expected behavior for a run that has already converged, not a malfunction.

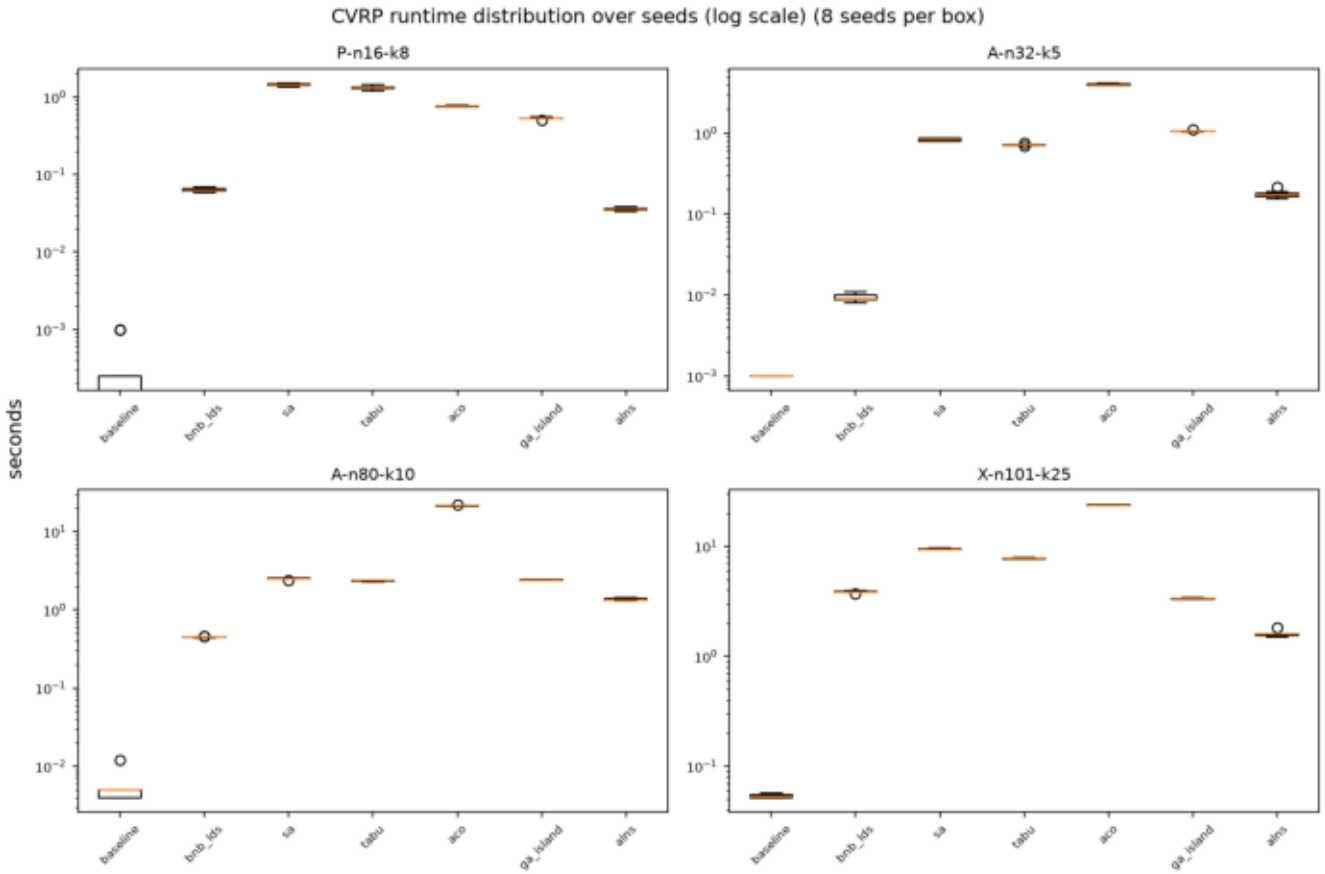
4.3 Seed robustness (8 seeds)

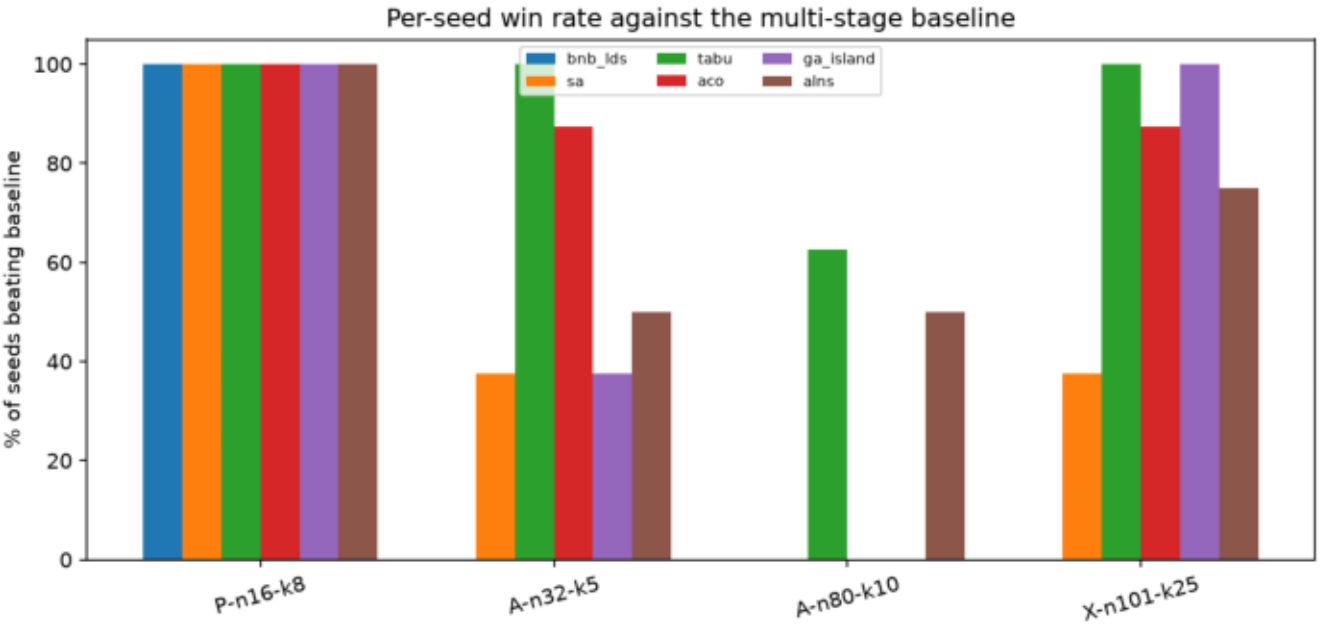
Best values can hide luck, so the effective algorithm set was rerun with eight seeds (42–49) on four representative instances — 224 runs, all feasible. This robustness study was measured with the Stage 10 configuration, before the advanced local-search pass was added, so its distributions describe the tuned solvers; the final best values in the tables above come from the later rerun. The boxplots show the full gap distribution, not just the best:



Median-level reading, which is more honest than best-level:

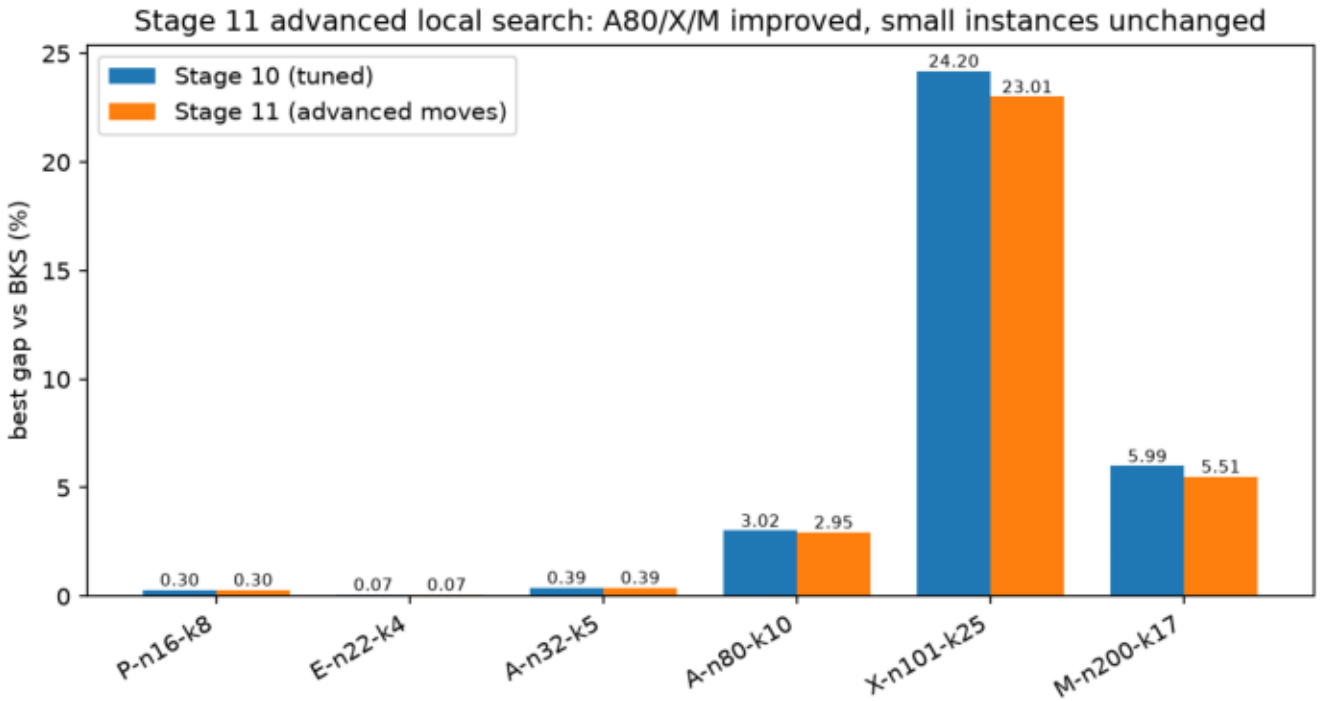
- P-n16-k8: everything beats the baseline on every seed; B&B/LDS is the most robust of all (0.30% on all 8 seeds — it is deterministic), and SA/ALNS have medians at 0.30% with occasional 0.43% seeds.
- A-n32-k5: the headline 0.39–0.41% results of ALNS and SA are seed-sensitive — their medians sit at the baseline 5.70%, meaning most seeds do not escape it. ACO is quietly the most consistent improver there (median 4.58%).
- A-n80-k10: ALNS improves on most seeds (median 4.90% in this pre-advanced study); Tabu improves slightly and consistently (median 4.89%); everything else sits on the baseline — the advanced pass later moved GA off the baseline here.
- X-n101-k25: GA-Island is the most robust improver (median 25.01% in this study); ALNS helps on some seeds only (median 26.86%).





The win-rate chart summarizes the same story: on the small instance every method wins every seed; on the tighter instances only some methods win reliably. With 8 seeds these are descriptive statistics — no significance test is claimed — but they make one thing clear that the best-value tables cannot: ALNS's advantage is broad but not uniform, and some of the tuned wins (especially on A-n32-k5) depend on the seed.

4.4 Advanced local-search impact (Stage 11)



The advanced moves were validated on scaled budgets before acceptance and then confirmed by the full final rerun. Against the previous committed final results, the project-best gaps moved as follows:

instance	before advanced	after advanced	change
A-n80-k10	3.0168%	2.9466%	?0.07
X-n101-k25	24.1950%	23.0063%	?1.19
M-n200-k17	5.9946%	5.5139%	?0.48
P-n16-k8 / E-n22-k4 / A-n32-k5	—	—	unchanged (gate off)

There are no project-best regressions. GA-Island gained the most: its memetic polish improved all three gated instances (A-n80 4.95?4.66, X-n101 24.20?23.01, M-n200 8.43?8.01) and it now beats the baseline on 6/6. ALNS improved on A-n80-k10 (3.02?2.95) and M-n200-k17 (5.99?5.51), but its enhanced variant regressed on X-n101-k25 (25.01?25.88) — the intensification restarts the search from polished bests, and on this extremely tight instance that particular trajectory change hurt. The regression is contained (the X project-best still improved through GA) and reported rather than tuned away. To be clear, X-n101-k25 is not solved: 23.01% remains a large gap, for the capacity-packing reasons in the note below.

Honest note on X-n101-k25: its total demand is 5147 while the total fleet capacity is 25 × 206 = 5150 — only 3 units of slack over 25 routes, so almost every route must be loaded completely full. Clarke-Wright needed 28 routes there, and a subset-sum packing repair (Section 5) was required just to reach feasibility. That packing ignores geometry, and with nearly zero capacity slack the usual local moves (relocate, cross-route swap) are almost always capacity-infeasible, so the metaheuristics could barely improve the start. The advanced inter-route moves recovered some of it, but the remaining 23.01% gap is real and reported as such.

5. Multi-stage CVRP Heuristic

The baseline heuristic runs in five stages:

1. Construction with Clarke-Wright savings (merge route ends by descending saving under the capacity limit).
2. 2-opt inside each route (reverse inner segments while improving).
3. Relocate between routes (move single customers while improving).

4. Vehicle-count repair: first empty surplus routes and reinsert their customers at cheapest feasible positions; if that fails, deterministic rebuilds (cheapest insertion, best-fit packing, sweep by polar angle); as a last resort a subset-sum packing that fills vehicles one at a time as full as possible over the integer demands — this is what makes X-n101-k25 feasible.
5. Final validation — a failed repair is returned as feasible=False with errors, never hidden.

Why the repair stage was necessary: Clarke-Wright merges routes only while the merge is profitable, so on instances with a tight fleet it can stop with more routes than vehicles exist. This actually happened twice in the final runs — P-n16-k8 (9 routes for 8 vehicles) and X-n101-k25 (28 routes for 25). Without the repair, every algorithm that starts from the baseline would inherit an infeasible incumbent, which is exactly what the first final run showed before stages 8-B2/9-B2. The subset-sum packing below is the piece that finally made X-n101-k25 feasible:

src/cvrp/local_search.py — subset-sum vehicle repair

```
636 | def build_routes_subset_sum_packing(instance, customer_order):
637 |     """Fill vehicles one at a time as full as possible, using a subset-sum
638 |     table over the integer demands.
639 |
640 |     Needed when total demand almost equals total capacity: X-n101-k25 has
641 |     5147 demand against 25 * 206 = 5150 capacity, so almost every route must
642 |     be loaded completely full, which greedy packing cannot find.
643 |     Returns customer groups or None.
644 |     """
645 |     capacity = int(instance.capacity)
646 |     if instance.capacity != capacity:
647 |         return None  # table needs integer capacity
648 |     demands = {}
649 |     for customer in customer_order:
650 |         demand = instance.demands.get(customer, 0.0)
651 |         if demand != int(demand) or demand > capacity:
652 |             return None  # table needs integer demands
653 |         demands[customer] = int(demand)
654 |
655 |     remaining = list(customer_order)
656 |     total_left = sum(demands[c] for c in remaining)
657 |     groups = []
658 |     for bins_left in range(instance.vehicle_count, 0, -1):
659 |         if not remaining:
660 |             break
661 |         # everything not in this bin must still fit into the other bins
662 |         lower = max(0, total_left - (bins_left - 1) * capacity)
663 |         if lower > capacity:
```

The idea is simple to state: when total demand almost equals total fleet capacity, almost every vehicle must leave completely full, so the repair fills vehicles one at a time with a subset of customers whose demands sum as close to the capacity as possible (a small dynamic-programming table over the integer demands), with a lower bound making sure the remaining customers still fit into the remaining vehicles.

Complexity: everything here is heuristic, not exact. One 2-opt pass over a route of length L evaluates $O(L^2)$ candidate reversals; since the Stage 11 performance pass each candidate is priced in O(1) from its four changed edges (delta = d[a][c] + d[b][d] ? d[a][b] ? d[c][d]) instead of recomputing the whole route, which made the pass 7–136× faster (route length 10–200 in the microbenchmark) while producing byte-identical routes. A NumPy distance matrix was also benchmarked and rejected: scalar indexing into an ndarray inside these Python loops measured about 4×

slower than plain lists. The relocate pass scans all customer/position pairs, roughly $O(n^2)$ per pass, and can optionally be restricted to k-nearest candidate lists. The repair adds packing work (the subset-sum table is $O(n \cdot \text{capacity})$ per vehicle) but guarantees the route count fits the fleet, which turned out to be essential on two of the six official instances.

6. CVRP Algorithms

All six start from the multi-stage baseline solution, share one random relocate/swap/2-opt neighborhood where applicable, and were run with the same per-instance budget and timeout for a fair comparison.

6.1 Simulated Annealing

Full solutions as states; one random neighbor per iteration; accept improvements always and worse candidates with probability exp(-delta/T). The first run used an untuned schedule (T?=100, cooling 0.995, iterations = plan budget) and mostly stayed near the baseline (mean best gap 8.50%). The tuning pass exploited how cheap one SA iteration is compared with the other methods: 50× more iterations, cooling 0.9995 and T?=500 — still well inside the shared per-instance timeout. That alone dropped SA's mean best gap to 6.74% and made it the best method on P-n16-k8 (0.30%) and E-n22-k4 (0.07%). It still cannot beat the baseline on A-n80-k10 or M-n200-k17.

6.2 Tabu Search

Samples 40 candidates per iteration from the shared neighborhood, forbids recently visited solutions via a customer-sequence signature (FIFO tenure 30), with aspiration on a new global best. Tied for the best P-n16-k8 result (0.43%) and was solid on the small instances; mean gap 8.12%.

6.3 Ant Colony Optimization

Ants build routes customer by customer with probability proportional to pheromone^? · (1/distance)^? under the capacity limit, light 2-opt per ant, evaporation plus deposits on the iteration and global best. ACO produced the best feasible X-n101-k25 result of the first full run (25.45%, since surpassed by the tuned GA) — its constructive nature sidesteps the frozen-local-moves problem there — at the price of the highest runtime (mean 32.8 s per run).

6.4 GA Island Model

Giant-tour chromosomes (customer permutations) split into routes by a capacity-aware scan; OX crossover, swap/inversion mutation, tournament selection, elitism 1; two islands with ring migration every 20 generations. The tuned final settings raise the population from 12 to 30 and the mutation rate from 0.15 to 0.3 (tuning also exposed and fixed an initial-population bug that looped forever when the population was larger than the number of distinct permutations). Stage 11 added a size-gated memetic step: every 10 generations the best solution is polished with the advanced inter-route pass and its chromosome is reinjected into island 0. Result: mean best gap 6.96%, beats the baseline on 6/6 instances, holds the overall best X-n101-k25 result (23.01%), and finally moved off the baseline on A-n80-k10 (4.66%).

6.5 ALNS

Destroy operators and repair operators chosen by adaptive weights, with simulated-annealing acceptance. The basic pool is random/worst removal plus greedy/regret-2 insertion; the enhanced pool adds Shaw-style related removal, full-route removal, segment removal and regret-3 insertion. The final rerun executed both variants on every instance, and the reported "alns" result follows the rule declared before the rerun: enhanced

everywhere except M-n200-k17, where validation had shown a +1.02% regression, so the basic variant is used there. Stage 11 added a size-gated intensification: when ALNS finds a new best on an instance with at least 60 customers, the best is polished with the advanced inter-route pass (relocate/swap/Or-opt/2-opt*, candidate lists k=10) and the search continues from it, plus one final polish before returning. With that policy ALNS remains the strongest method overall: lowest mean best gap (5.85%), beats the baseline on 6/6 instances, and holds the best result on A-n32-k5 (0.39%), A-n80-k10 (2.95%) and M-n200-k17 (5.51%) at a moderate runtime. Its one disclosed weak spot: the intensification worsened the enhanced variant's X-n101-k25 trajectory (25.01?25.88, Section 4.4).

6.6 Branch-and-Bound / LDS

A time-limited search over customer insertion decisions (hardest customers first), where taking the k-th cheapest insertion costs k discrepancy, with a partial-cost bound against the incumbent. With the original shallow budget (discrepancy 3) it never beat its starting incumbent anywhere. The final version adds a small-instance mode: instances with at most 25 customers get discrepancy 15 and a 2M-node cap (still timeout-capped, still falling back to the incumbent). Measured effect: P-n16-k8 improves from 2.65% to 0.30% in ~0.03 s and E-n22-k4 from 3.34% to 0.07% in ~1.1 s — both tie the best results any metaheuristic found, and being a deterministic search, it hits them on every seed. The same depth was tested on A-n32-k5 (31 customers, depth 12, 8 s) without any improvement, so the threshold honestly stops at 25 customers; on the four larger instances B&B/LDS still returns its incumbent (overall mean gap 7.77%, beating the baseline on 2/6). It remains exact-inspired, not a full exact solver for these sizes.

6.7 Iterated Local Search (ILS) — the explicit iterated driver

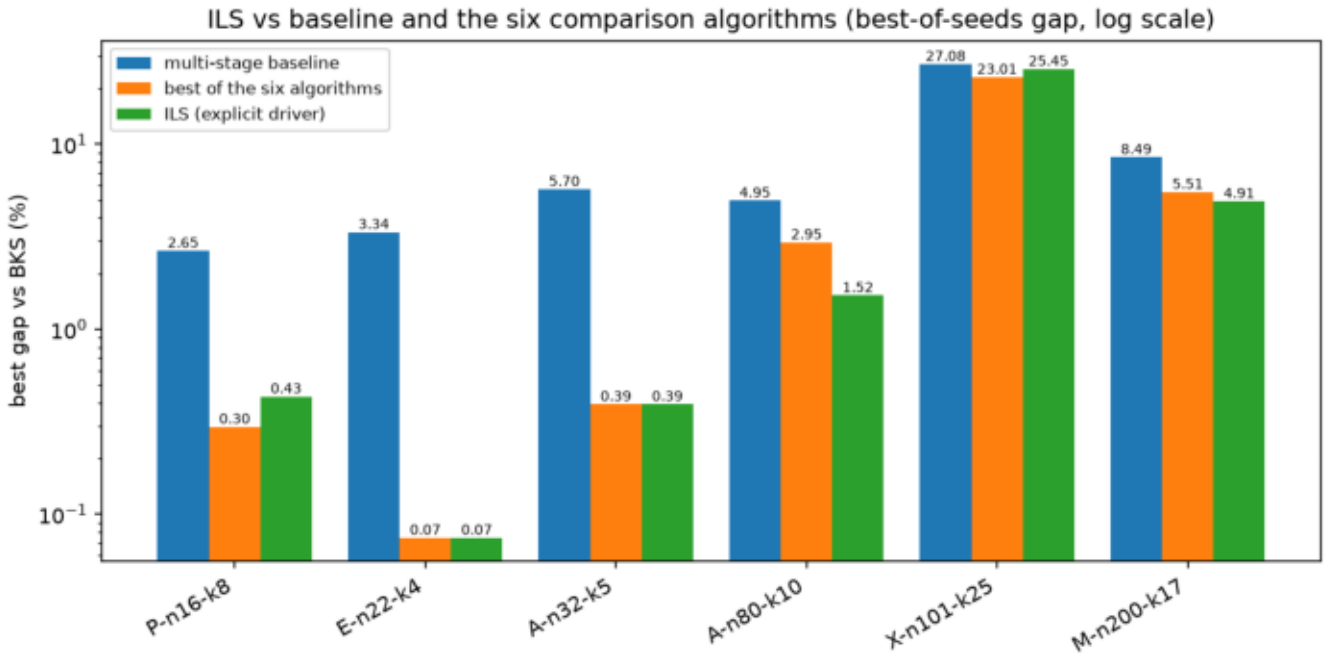
The lecture framed the meta-heuristics as instances of an abstract iterated local search; this section is the explicit implementation of that requirement (src/cvrp/solvers/ils.py, Stage 13-A). ILS is not a rebrand of SA, Tabu or ACO — those walk one random neighbor at a time, while ILS jumps between complete local optima, and its module imports none of the other solvers (a dedicated test enforces this). One ILS iteration has the classic four ingredients:

1. Initial solution: the multi-stage baseline (Section 5), improved to a first local optimum by the deterministic local-search stack.
2. Perturbation (exploration): a ruin-and-recreate kick removes a random subset of customers — the kick strength, ~10% of the customers (2 to 30) — and greedily reinserts each at its cheapest capacity-feasible position in random order. Rebuilding several routes at once deliberately throws the search out of the current local optimum, far beyond any single relocate/swap/2-opt step. The kick only reuses existing route slots, so the vehicle limit is never violated; if the greedy repair cannot place every removed customer, the kick is rejected and the previous solution is kept (never an infeasible state).
3. Local search (exploitation): the strongest deterministic stack in the project — intra-route 2-opt plus inter-route relocate, swap, Or-opt and 2-opt*, with k = 10 candidate lists on instances with at least 60 customers (the same gating the other solvers use).
4. Acceptance criterion (at the ILS level): an improved local optimum is always accepted; a worse one is accepted only within a 2% relative threshold (threshold acceptance) — this is the exploration/exploitation dial: 0% would be a pure descent that stalls in the first good basin, a large threshold would degenerate toward a random restart. After 20 iterations without a new global best, the search restarts from the best solution and the kick strength grows by one (capped at 2× the base), widening exploration instead of looping. Every accepted candidate and the final best are validated for feasibility.

Results (same plan as the six algorithms: seeds 42/43/44, per-instance budgets 300–1200 iterations and timeouts 30–300 s; evidence snapshots report/evidence/cvrp_ils_runs.csv, cvrp_ils_summary.csv,

cvrp_ils_manifest.json; 18/18 runs feasible):

instance	best cost	best gap	mean gap	std cost	mean elapsed (s)	mean CPU (s)
P-n16-k8	451.95	0.4327%	0.4327%	0.00	0.15	0.15
E-n22-k4	375.28	0.0746%	0.0746%	0.00	0.13	0.13
A-n32-k5	787.08	0.3931%	0.3931%	0.00	0.52	0.52
A-n80-k10	1789.83	1.5219%	1.8268%	5.34	6.16	6.16
X-n101-k25	34613.14	25.4508%	26.3938%	225.65	0.69	0.69
M-n200-k17	1337.57	4.9072%	5.4776%	9.24	64.81	64.80



The figure compares, per instance (log scale, lower is better), the best-of-seeds gap of the multi-stage baseline, the best of the six comparison algorithms, and ILS. ILS's mean best-of-seeds gap over the six instances is 5.463% (mean over all seeds 5.766%) — below the best of the six (ALNS, 5.851%). On A-n80-k10 (1.5219% vs 2.9466%) and M-n200-k17 (4.9072% vs 5.5139%) ILS holds the new overall project-best results: the compound kick plus the full deterministic re-optimization reaches basins the one-move walkers never enter. On E-n22-k4 and A-n32-k5 it exactly ties the project best, and on P-n16-k8 it stops at 0.4327% (vs 0.2967%): every seed hits the same local optimum there, and the 2%-threshold acceptance never wanders far enough on a 15-customer instance where one customer is 1% of the solution.

Honest X-n101-k25 note: ILS reaches 25.4508% — better than the baseline (27.08%) but behind GA-Island (23.0063%). The cause is the same capacity-packing effect from Section 4.4: with 3 units of fleet slack the greedy reinsertion almost never finds a feasible slot for a removed customer, so most kicks are rejected and ILS burns its full 1000-iteration budget in under a second without real exploration. A packing-aware kick (ejection-chain style) is the natural next step and is listed as future work. Note also that on A-n80-k10 and M-n200-k17 the iteration budget, not the timeout, was the binding limit (6 s of 120 s and 65 s of 300 s used) — the comparison stays fair because ILS received exactly the same budget and timeout values as the six algorithms.

Complexity: ILS ? iterations × (perturbation + local search). The kick costs O(strength · n^2) for its cheapest-insertion scans; the local-search stack dominates with roughly O(n^2) per best-improvement pass (O(1) delta per candidate move, optionally pruned by candidate lists). The perturbation adds exploration overhead per iteration, but it is exactly what lets the search escape local optima that the deterministic stack alone cannot leave.

7. Ackley Adaptations

SA and Tabu Search are natural continuous methods here (Gaussian steps; a tabu list of rounded points). GA-Island and ALNS operate directly on continuous vectors (blend crossover and Gaussian mutation; dimension-wise destroy and repair). ACO and B&B/LDS are honest discretizations (pheromone over bins per dimension; limited-discrepancy search over bin choices) — the canonical versions of these methods are combinatorial, and the report does not claim otherwise. Random search is only a sanity baseline. The results in Section 3 reflect exactly this: the methods whose adaptation points toward the origin (ALNS's toward-zero repair, B&B/LDS's zero-first bin order) reached it, the general-purpose ones (GA-Island, Tabu) got close, and untuned SA performed worst.

8. Part B — Rush Hour with GP and GEP

A* solves 6×6 Rush Hour boards with g = number of moves and a pluggable heuristic h. GP and GEP evolve h as expressions over three board features — distance of the red car to the exit, number of blocking cars, number of free exit cells — plus small constants, with protected operators (+, -, *, /, min, max, abs, neg, log). Fitness strongly rewards solved puzzles (10000 each) and penalizes expanded nodes, solution cost, timeouts and node-cap hits. Every evaluation runs under the safety caps from Section 2.3.

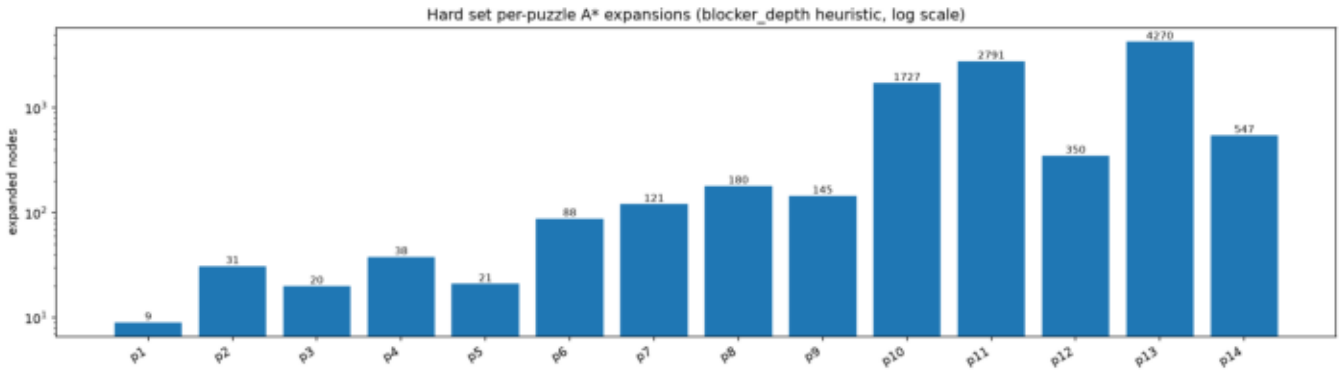
GP uses expression trees with subtree crossover and mutation. GEP is a separate framework: a linear genome with a head (functions or terminals) and a tail (terminals only), decoded as a Karva K-expression; its operators are point mutation and one-/two-point crossover on the flat gene string. The decoder is the heart of GEP — it reads the linear genome left to right and attaches children level by level, ignoring leftover symbols:

src/gep/decoder.py — Karva decoding

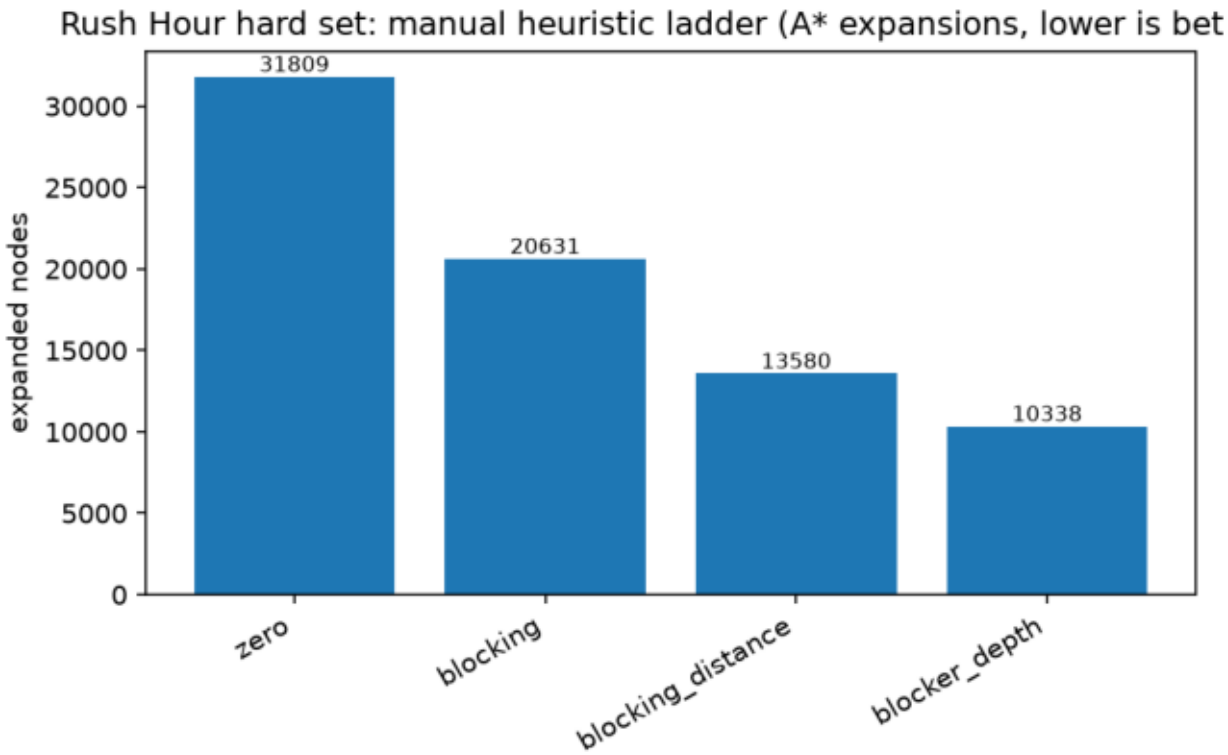
```
27 | def decode_genome_to_tree(genome: GEPGenome) -> GPTree:
28 |     symbols = genome.genes
29 |     root = _make_node(symbols[0])
30 |     open_nodes = deque()
31 |     if ARITY.get(symbols[0], 0) > 0:
32 |         open_nodes.append((root, ARITY[symbols[0]]))
33 |
34 |     index = 1
35 |     while open_nodes:
36 |         node, needed = open_nodes[0]
37 |         if len(node.children) == needed:
38 |             open_nodes.popleft()
39 |             continue
40 |         if index < len(symbols):
41 |             symbol = symbols[index]
42 |             index += 1
43 |         else:
44 |             symbol = "0.0"  # genome ended early, pad with a safe terminal
45 |         child = _make_node(symbol)
46 |         node.children.append(child)
47 |         arity = ARITY.get(symbol, 0)
48 |         if arity > 0:
49 |             open_nodes.append((child, arity))
50 |
51 |     return GPTree(root=root)
52 |
53 |
```

8.1 The hard benchmark (main Part B evaluation)

The first comparison used a 4-puzzle evaluation set, and everything — including plain manual heuristics — solved all of it with 4 expanded nodes, so GP and GEP tied perfectly at fitness 39956. That result is kept only as a historical smoke check; the main evaluation is a hard Rush Hour benchmark of 14 puzzles committed under examples/rushhour_hard_eval.txt. Every puzzle was verified with the project parser and solved by the project A* before being committed, with optimal solutions of 4–17 moves and measured difficulty spanning 36 to 13,243 zero-heuristic expansions.



A ladder of manual heuristics anchors the comparison (identical caps for everything: 15,000 nodes and 2 s per puzzle, 20 s total per evaluation):



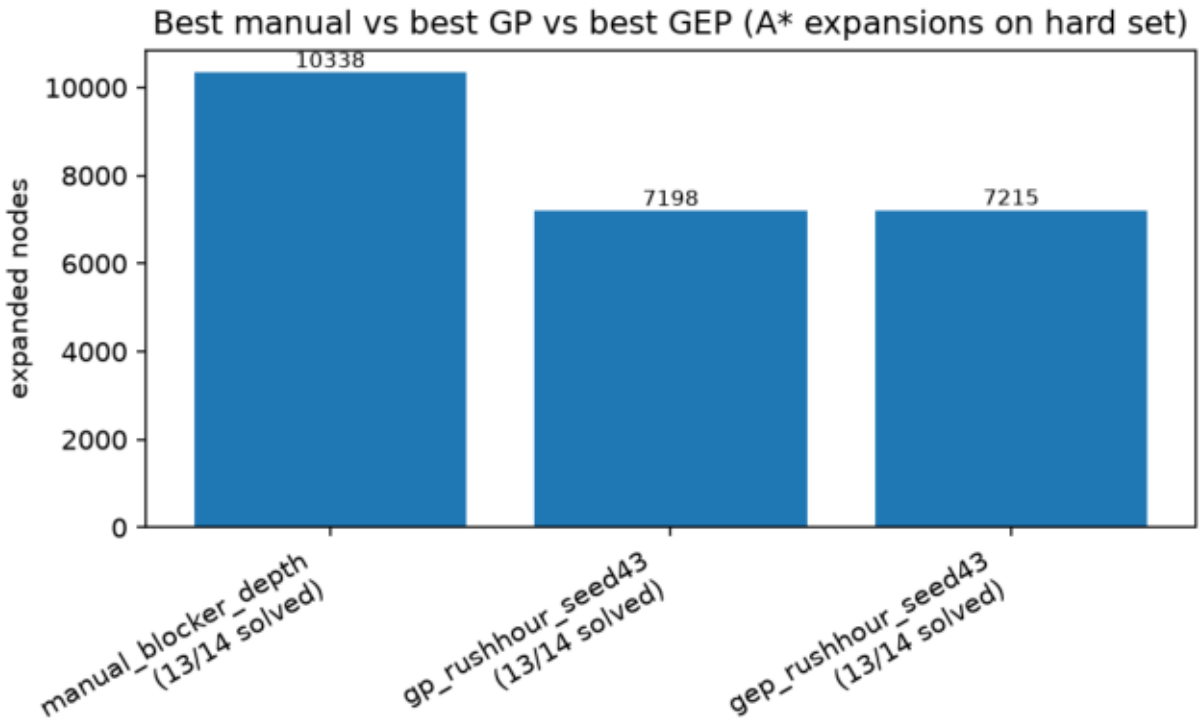
Each added ingredient pays off: zero solves only 9/14 within the caps (31,809 expansions), blocking gets 13/14 (20,631), blocking+distance needs 13,580, and blocker_depth — which also penalizes blockers that are themselves stuck — needs only 10,338. That monotone ladder is what makes the set a meaningful benchmark.

The same ladder ranks the manual heuristics on all three assignment axes —

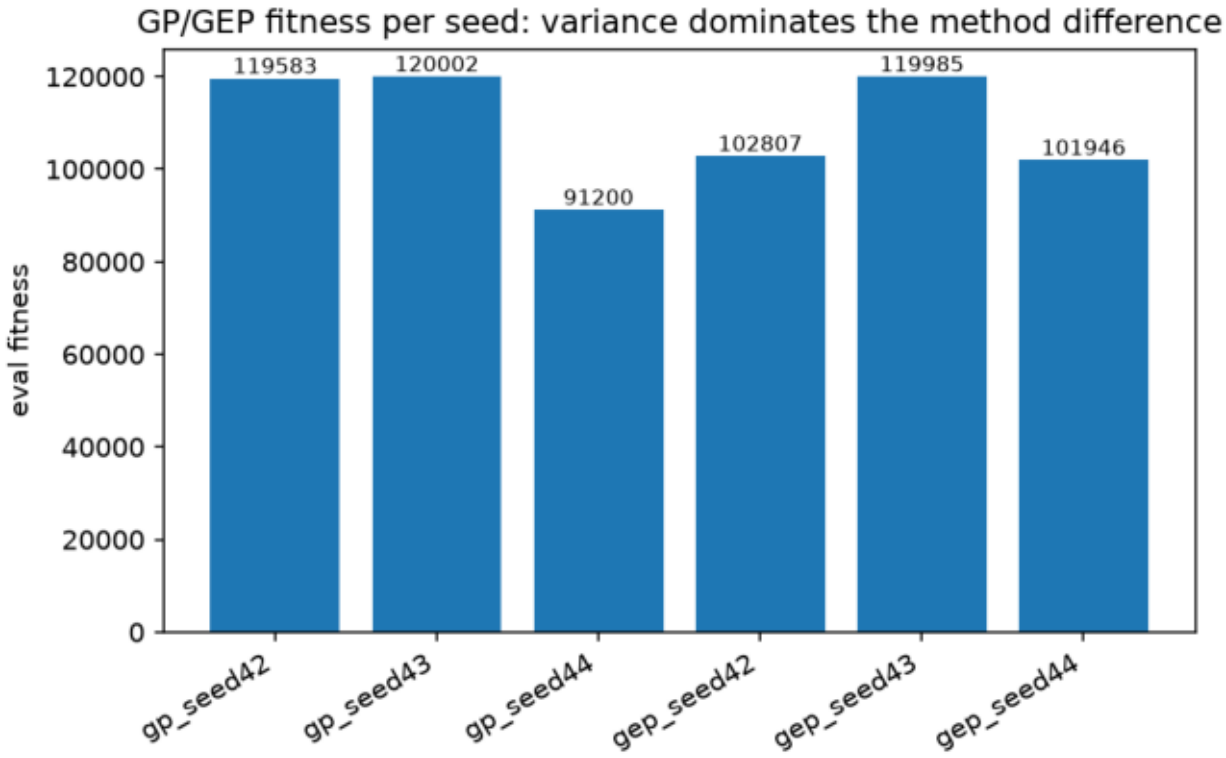
quality, computational complexity, and time to solve (total elapsed/CPU over the 14 puzzles, from rushhour_hard_manual_summary.csv): zero (9/14 solved, 11.85 s elapsed / 11.85 s CPU) is O(1) per state but wastes A* expansions; blocking (13/14, 8.40 s) adds one O(cars) scan of the exit row; blocking_distance (13/14, 5.89 s) adds the red-car distance at the same O(cars) cost; and blocker_depth (13/14, 5.57 s) pays a full legal-move generation per state to spot blockers that are themselves stuck — the most expensive evaluation per state, yet the fastest overall because better guidance saves thousands of expansions. The ranking lesson is explicit: a slightly costlier, more informed heuristic wins on wall clock, not just on node counts.

GP and GEP were then run with identical budgets (population 30, 20 generations, seeds 42–44; training on the 4 original puzzles plus the 3 easiest hard ones; evaluation on all 14):

run	eval fitness	solved	expanded	best expression
gp seed 42	119583	13/14	7617	min((min(blocking, blocking) * (distance - 0)), ...)
gp seed 43	120002	13/14	7198	(abs((blocking * (blocking * blocking))) / abs(0.5))
gp seed 44	91200	11/14	12150	(((distance * distance) / free) ... * (blocking / 0.5))
gep seed 42	102807	12/14	12533	(max(max(1, 0), 5) - ((free - distance) * 2))
gep seed 43	119985	13/14	7215	max((5 * blocking), ((1 + distance) * blocking))
gep seed 44	101946	12/14	13384	(blocking + (distance / (abs(blocking) - free)))
manual blocker_depth	116852	13/14	10338	(hand-written)



The core finding: the best evolved heuristics from both frameworks beat the strongest manual heuristic — GP 120002 and GEP 119985 vs blocker_depth 116852 — solving the same 13/14 puzzles with about 30% fewer A* expansions (?7,200 vs 10,338). Evolution genuinely found better guidance than the hand-written baselines here.



GP vs GEP stays an honest tie at the top: 120002 vs 119985 is a 0.01% difference, while the per-seed spread within each method is ~30,000 fitness points (GP's worst seed drops to 91200, GEP's to 101946). Seed variance dominates the method difference, so no winner is declared. Expression diversity is 1.00 for both. Production (evolution) time does separate them: on the hard benchmark one GP run takes 60.3 s on average (65.1/67.4/48.5 for seeds 42–44) against GEP's 44.0 s (55.3/30.3/46.3), from rushhour_hard_gp_gep_runs.csv; the small 4-puzzle comparison shows the same ratio (GP 1.21 s vs GEP 0.77 s mean). GEP's flat point/one-/two-point operators on a fixed-length gene string are simply cheaper than GP's subtree crossover and mutation, and its Karva decoding keeps expressions shorter — the same parsimony that shows up in its more compact best expressions. Low computational complexity is rewarded in both fitness designs, as the assignment asks: in the A*-guided fitness implicitly — every candidate runs under a 2 s per-puzzle time cap, so an expensive-to-evaluate expression times out and loses 2000 fitness per timeout — and in the direct-planner fitness (Section 8.2) explicitly, with a parsimony term of ?2 per expression node. Notably, no run solves 14/14 under the caps — the hardest puzzle needs more than the 15,000-node budget with any evolved or manual guide — which means the benchmark still has headroom and the results are not saturated like the old 4-puzzle set. (Numbers this close to the time caps can shift by a fraction of a percent between runs; the ordering has been stable.)

8.2 Bonus: direct GP/GEP Rush Hour planner without A*

The main Part B method evolves heuristics that are used *inside* A*. The assignment's bonus asks the opposite question: can GP/GEP act directly as the planner, with no A* at all? This subsection is that exploratory bonus — it does not replace the A*-guided result above.

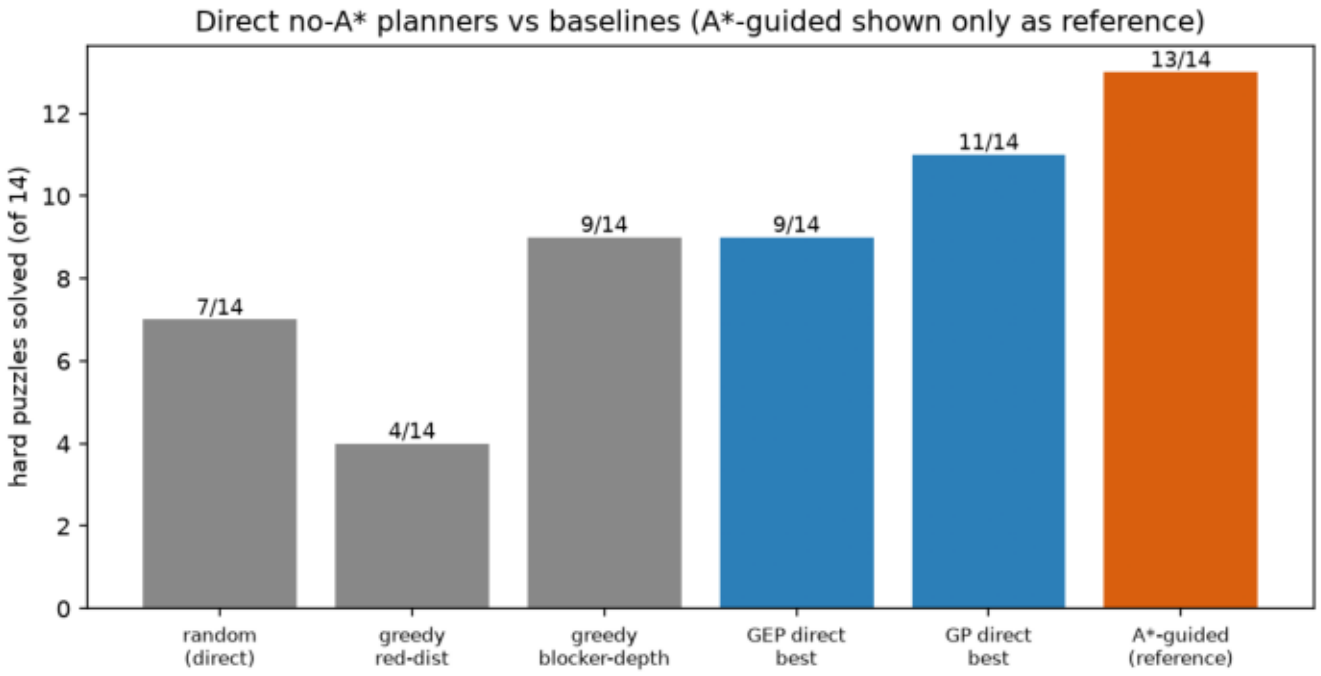
How the direct planner works. A policy expression is rolled out greedily: at each state the planner enumerates all legal moves, scores every move with the evolved expression (lower = better), applies the best one with a deterministic tie-break, and stops on solved, at 120 steps, or at a 2-second timeout. There is no open list, no f = g + h, no backtracking — and a visited-state set makes the policy prefer moves to

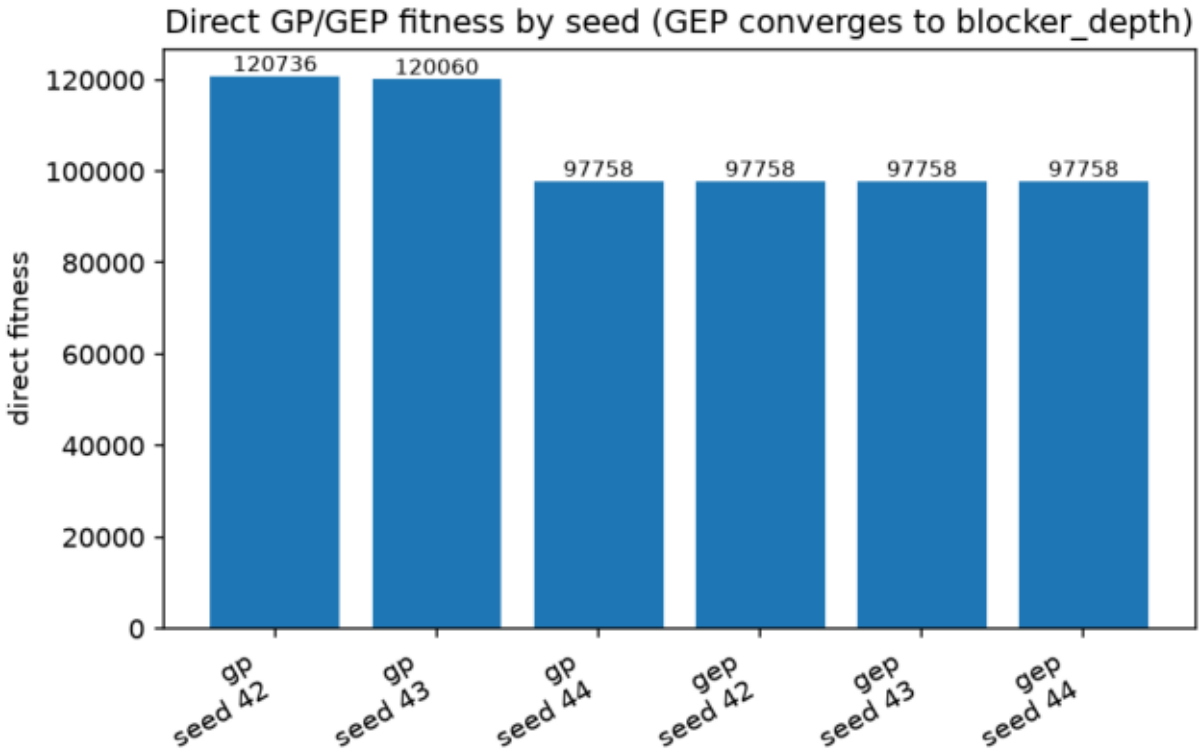
unvisited states (revisits are counted and penalized in fitness). The code is src/rushhour/direct_planner.py; a test suite monkeypatches the A* solver to raise if it is ever called during direct planning.

Move features. Instead of the three state features the A*-heuristics use, direct policies see eleven features of each candidate move: red-car distance and blockers/blocker-depth/free-exit cells *after* the move, the deltas of distance and blockers caused by the move, whether the moved vehicle is the red car and whether it moves toward the exit, the move length, the mobility (number of legal moves) of the resulting position, and a visited indicator for cycle avoidance (src/rushhour/direct_features.py).

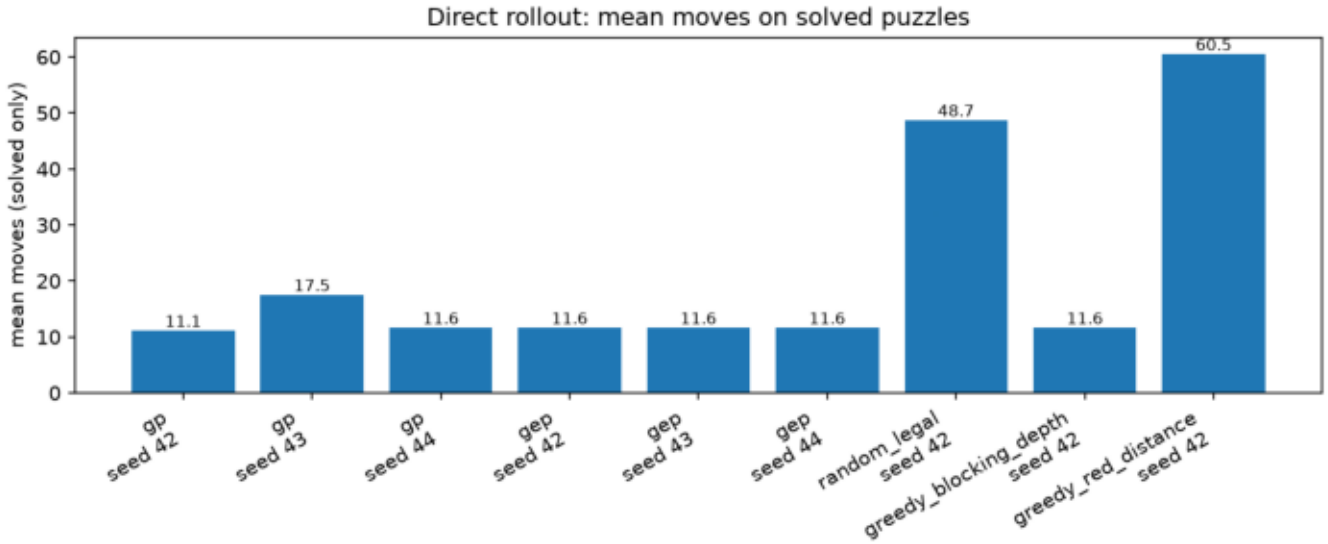
Benchmark. Same 14 hard puzzles, seeds 42–44, population 30, 20 generations, identical rollout caps for every method. Three hand-written direct policies run under the *same* rollout as baselines, so the comparison is direct-vs-direct, not direct-vs-A*:

direct method	solved (best)	notes
random legal move	7/14	seeded random walk
greedy red-distance	4/14	oscillates (91 cycle fallbacks)
greedy blocker-depth	9/14	strongest manual direct policy
GEP direct	9/14 (all seeds)	converges to blocker_depth, size 1
GP direct	11/14 (seeds 42, 43; mean 10.3)	composite expressions, sizes 22/10/1
A*-guided evolved heuristic (reference)	13/14	not direct — shown for scale





GP's best direct policies (fitness 120736 and 120060, expression diversity 0.77–0.80) combine blocker_depth with mobility and red_distance terms and solve 11/14 — a real improvement over the best hand-written direct policy (9/14). GEP is honestly less interesting here: all three seeds converged to the single terminal blocker_depth (fitness 97758, size 1), exactly rediscovering the best manual feature and tying it, never beating it. The parsimony pressure that kept GEP compact in the A* benchmark works against it in this richer feature space.



Honest conclusion. The A*-guided approach remains clearly stronger (13/14 vs 11/14): A* has search memory and frontier management, while the greedy rollout commits to one move per state and can only avoid cycles, not recover from them. Per step, however, the direct planner is very cheap — it scores only the current legal moves (about a dozen boards) instead of expanding thousands of A* nodes, and solved puzzles take it 11–18 moves on

average. The bonus stands as a positive exploratory result: evolution found a better direct policy than we could write by hand, and the no-search planner solves 11 of 14 measured-hard puzzles — but the main Part B result stays the A*-guided benchmark of Section 8.1.

9. Results

The main tables and figures are in Sections 3 (Ackley), 4 (CVRP) and 8 (GP/GEP); the committed report figures live under report/figures/ and small evidence snapshots under report/evidence/. The raw per-run rows live under results/final_experiments/raw/ (144 CVRP rows including both ALNS variants, 21 Ackley rows) plus the hard Rush Hour benchmark under results/final_experiments/rushhour_hard/ and the ILS evidence under results/final_experiments/ils/; the aggregated tables are under results/final_experiments/summary/. The execution manifest (final_execution_manifest.json) records what ran, with which budgets, the tuned settings, and the pre-declared ALNS policy.

The compact CVRP statistics below aggregate the per-instance summary rows over the six instances (policy view for ALNS; ILS from its own evidence run under identical seeds/budgets/timeouts). "Mean best gap" averages the best-of-seeds gap per instance; "mean gap" averages over all seeds; "std cost" is the mean per-instance standard deviation of the solution cost across seeds (0.00 = every seed found the same cost); elapsed is wall clock and CPU time is process time, both averaged per run. Every single run of every algorithm was feasible (18 runs each: 6 instances × 3 seeds).

algorithm	mean best gap	mean gap	std cost	elapsed (s)	CPU time (s)	feasible
ils (explicit driver)	5.463%	5.766%	40.04	12.08	12.08	18/18
alns (policy + advanced)	5.851%	7.029%	40.10	4.72	4.58	18/18
sa (tuned)	6.740%	7.430%	25.30	3.48	3.40	18/18
ga_island (tuned + advanced)	6.958%	7.102%	35.49	2.77	2.72	18/18
aco	7.558%	7.739%	43.88	24.13	23.49	18/18
bnb_lds (small-instance mode)	7.765%	7.765%	0.00	1.36	1.32	18/18
tabu	8.092%	8.120%	0.42	3.16	3.10	18/18
baseline	8.703%	8.703%	0.00	0.02	0.02	18/18

Elapsed and CPU means are nearly identical for every method — the solvers are single-threaded pure Python with no I/O in the hot loop, so wall clock ? CPU time; only ACO shows a visible difference (24.13 vs 23.49 s), i.e. a small fraction of its wall clock is spent off-CPU. The deterministic methods (baseline, B&B/LDS) and near-deterministic Tabu have (near-)zero std; the stochastic improvers pay seed variance for their better means.

10. Analysis and Discussion

- CVRP pattern. Gaps grow with instance size: 0.30% or less on P-n16-k8 and E-n22-k4, 0.39% on A-n32-k5, ~3–5.5% on A-n80-k10 and M-n200-k17. ALNS keeps the lowest mean best gap (5.85%) and, together with the advanced GA-Island, beats the baseline on all six instances.
- What tuning and the advanced moves bought (and did not). Tuning: P-n16-k8 0.43?0.30, A-n80-k10 4.03?3.02, X-n101-k25 25.45?24.20. The Stage 11 advanced moves then pushed the large instances further: A-n80-k10 ?2.95, X-n101-k25 ?23.01, M-n200-k17 5.99?5.51, with the small instances unchanged by the size gate and no project-best regressions. The tuned SA improvement came purely from spending its unused time budget on more iterations — a fairness-preserving change, since all methods share the same timeout. The enhanced ALNS improvement needed real new operators (Shaw/route/segment removal, regret-3). Its M-n200-k17 regression was handled by the pre-declared hybrid policy, and the one advanced-move regression (enhanced ALNS on X, 25.01?25.88) is disclosed in Section 4.4, not hidden.
- X-n101-k25. With 3 units of total capacity slack, feasibility is a bin-packing problem and single-customer local moves are nearly frozen: any relocate overfills a route. The Stage 11 compound moves (swap, Or-opt, 2-opt*) were added precisely for this and recovered about 1.2 points through GA (24.20?23.01), but the gap stays large. Full ejection

chains — the next escalation — stayed out of scope.
- Ackley. Unchanged by this stage. The warm-up separated the methods, but part of that separation comes from how each was adapted (Section 7). Untuned SA losing to random search there remains a useful reminder that parameter choices matter as much as the algorithm name — the CVRP side proved the same point in reverse when tuned SA jumped from 8.50% to 6.74%.
- GP vs GEP. On the hard benchmark both frameworks beat the strongest manual heuristic by ~30% fewer expansions, and they remain effectively tied at the top (120002 vs 119985) with per-seed variance ~30,000 fitness points — far larger than the method difference. GEP's best expressions stay visibly more compact. Both frameworks are genuinely different representations, not renames.
- ILS and the exploration/exploitation trade-off. The explicit ILS driver (Section 6.7) makes the trade-off visible as two separate dials: kick strength (how far exploration jumps) and acceptance threshold (how much exploitation is allowed to relax). With ~10% kicks and a 2% threshold it produced the project's best A-n80-k10 and M-n200-k17 results and the lowest mean best gap (5.463%), showing that restarting a strong deterministic local search from perturbed local optima can beat the adaptive one-move walkers. Its two failure modes are equally instructive: on P-n16-k8 the threshold is too tight relative to the instance's coarse cost granularity (it converges to the same local optimum on every seed), and on X-n101-k25 the kick itself is starved — the tight packing rejects nearly every greedy reinsertion, so exploration never actually happens.
- Runtime vs quality. ALNS delivers the best average CVRP quality at moderate runtime; ACO remains the most expensive; tuned SA uses its budget instead of leaving it idle. B&B/LDS with the small-instance mode ties the best results on the two smallest instances almost for free, but spends its budget without beating the incumbent everywhere else. The advanced moves bought their quality at a controlled cost: mean ALNS runtime on M-n200-k17 stayed flat (26.9?25.4 s basic, 22.0?22.5 s enhanced) and GA paid about 1.5 s extra per M-n200-k17 run, because the delta 2-opt speedup absorbed most of the added work.
- Seed robustness. The 8-seed analysis (Section 4.3, measured before the advanced pass) shows which wins are stable: ALNS is a broad but not uniform improver, GA is the reliable one on X, B&B/LDS is perfectly stable where it works, and the A-n32-k5 headline numbers are seed-lucky (median = baseline). Eight seeds give a descriptive picture, not a statistical significance claim.
- Limitations. One fixed budget/timeout profile per instance, three seeds for the main tables (eight for the robustness section, which predates the advanced pass), one tuning pass validated on the same six instances, an advanced-move regression on one algorithm/instance pair (Section 4.4), a B&B/LDS that is still ineffective beyond 25 customers, and a 14-puzzle Rush Hour set — the comparisons hold for this setup only, and no claim of class-leading performance is made without an external comparison.

10.1 Original engineering choices

Beyond the required algorithms, several design decisions in this project are our own engineering additions, all implemented and evidenced rather than proposed:

- Report-vs-evidence consistency gate — verify_report_matches_csv.py plus dedicated tests re-derive every headline number in this report from the committed evidence CSVs; the submission audit runs the gate, so a number cannot silently drift from its data.
- Subset-sum vehicle repair (Section 5) — the dynamic-programming packing that made X-n101-k25 feasible at all.
- Size-gated advanced local search (Section 4.4) — inter-route swap/Or-opt/2-opt* with candidate lists, activated by customer count after validation showed a small-instance regression.
- Ruin-and-recreate ILS kick with adaptive strength (Section 6.7) — restart acceptance that widens the kick on stagnation, which produced two new project-best results.
- Pre-declared ALNS policy (Section 4) — the enhanced/basic selection

rule was fixed before the final rerun, so no cherry-picking after the fact; both variants' raw rows are kept.
- Measured-difficulty hard Rush Hour benchmark and the no-A* direct planner bonus (Sections 8.1–8.2) — including the safety-capped evaluator that lets evolution run untrusted heuristic code safely.
- Smoke/final isolation and resumable final runner — smoke runs write to separate directories and the final suite skips finished raw CSVs, so partial reruns cannot contaminate the canonical evidence.
- Honest regression reporting — the ALNS-on-X and ALNS-on-M regressions are disclosed and analyzed instead of tuned away.

11. Complexity and Practical Considerations

- Feasibility checking is O(n) per solution and is run on every candidate the metaheuristics accept, plus once on every final result — the cost is small compared to neighborhood evaluation and it caught real bugs (route-count violations) that pure cost comparison would hide.
- Neighborhood costs. The shared relocate/swap/2-opt neighborhood costs O(n) per sampled neighbor (copy + cost); Tabu's 40 candidates per iteration make it ~40× SA per iteration, which matches the observed runtimes. The Stage 11 best-improvement passes (relocate, swap, Or-opt, 2-opt*) price each candidate move in O(1) from its changed edges and can prune their scans with k-nearest candidate lists; both the exact and the pruned neighborhoods are kept available, and k = 10 was accepted only after producing identical solutions to the full scan in validation.
- Stochastic variance. Three seeds per configuration; the summary CSVs report mean and standard deviation. The seeds are fixed in the plan, so every row can be regenerated exactly.
- Timeouts and fairness. All algorithms get the same timeout on the same instance, so slower-per-iteration methods are not silently given more work. Budgets differ per algorithm in meaning (iterations vs generations vs nodes), which is why the timeout is the binding fairness control on the large instances.
- ILS cost structure. One ILS iteration = one kick (O(strength · n^2) cheapest-insertion scans) + one full deterministic re-optimization (O(n^2)-per-pass best-improvement stack) — far heavier per iteration than SA's single sampled neighbor, which is why ILS runs hundreds of iterations where SA runs hundreds of thousands. The evidence shows the trade paid off on the two instances where deep re-optimization matters (A-n80-k10, M-n200-k17) and was wasted where the kick cannot act (X-n101-k25).
- No optimality claims. Everything here is heuristic or time-limited; the gaps against BKS quantify exactly how far the results are from the best known solutions.

12. Use of AI Tools

This declaration follows the course's policy on AI tools and code generators. AI tools were used substantially during development, as detailed below.

- Tool used: Claude (Anthropic), operated through the Claude Code CLI assistant, was used as a substantial aid across the project's staged workflow: planning the stage breakdown, writing and refactoring code, writing tests, debugging failures, reviewing consistency between the report and the evidence files, and polishing this report's text.
- What the AI did not do: no experimental number in this report was produced or edited by hand or by the assistant directly — every value comes from running the committed scripts on this machine, and the consistency gate (scripts/verify_report_matches_csv.py) plus the test suite re-check the report against the evidence CSVs on every run.
- Responsibility and understanding: the submitters reviewed all generated code, take full responsibility for it, and can explain every line, as the course requires. Code that could not be explained was rewritten until it could be.
- External code: no third-party or copied code is included beyond the declared open-source libraries in requirements.txt (numpy, matplotlib,

pandas, pytest); no unreviewed generated code was submitted.

13. Reproducibility

- Python 3.12.3, dependencies via pip install -r requirements.txt
  (numpy, matplotlib, pandas, pytest).
- Place the six official CVRPLIB files under data/official_cvrp/ and
  verify with python scripts/check_official_cvrp_data.py --strict.
- Smoke check: python scripts/run_smoke_suite.py --output-dir results/smoke_suite
- Print/validate the final plan:
  python scripts/print_final_experiment_plan.py --require-official-data
- Full final tuned run (resumable):
  python scripts/run_final_experiments.py --tuned-cvrp configs/tuned_cvrp_settings.json
  --rushhour-hard configs/rushhour_hard_benchmark.json
- Explicit ILS on one instance:
  python scripts/run_cvrp_ils.py --instance data/official_cvrp/A-n80-k10.vrp --iterations 800
  --seed 42 --timeout 120
- ILS evidence suite (same plan seeds/budgets/timeouts as the six):
  python scripts/run_ils_evidence.py --refresh-evidence
- Hard Rush Hour benchmark on its own:
  python scripts/run_gp_gep_hard_benchmark.py --puzzles examples/rushhour_hard_eval.txt --seeds 42
  43 44
- Direct no-A* planner bonus (Section 8.2):
  python scripts/run_gp_gep_direct_planner.py --seeds 42 43 44 --population 30 --generations 20
  (evidence snapshots: report/evidence/direct_*.csv and
  direct_planner_manifest.json)
- Old-vs-new comparison: python scripts/extract_final_results_v3.py
  (its output, final_v3_summary.txt, is the current final comparison
  summary and a copy is committed under report/evidence/)
- Evidence snapshots: python scripts/refresh_report_evidence.py copies
  and derives the small committed files under report/evidence/ from the
  local final results
- Report figures: python scripts/generate_report_figures.py, plus
  python scripts/generate_route_visualizations.py and
  python scripts/generate_convergence_figures.py, then
  python scripts/export_report_pdf.py for this PDF
- Audit (development repo): python scripts/audit_submission.py --check-results --check-pdf
- Audit (clean source package, no results/ needed):
  python scripts/audit_submission.py --submission-package or python a3.py audit-package
- Report facts: python scripts/extract_report_numbers.py
- Small evidence snapshots of the result files cited by this report are
  committed under report/evidence/ (summaries, GP/GEP runs, execution
  manifest). The full generated outputs stay under
  results/final_experiments/ locally and are not committed to Git, and
  the official .vrp files are user-provided data that is also kept out of
  the repository.

The final runner is what makes the experiments reproducible in practice:
it validates the plan and the official data before anything runs, executes
each part through the same Python functions the tests use, writes one CSV
row per run with its seed/budget/timeout, and skips already-finished raw
files — when the vehicle-count repair changed, only X-n101-k25 had to be
rerun while the other five instances were reused untouched:

src/experiments/final_execution.py — final suite

```
356 | def run_final_experiment_suite(plan_path="configs/final_experiment_plan.json",
357 |                                resume: bool = True, run_cvrp: bool = True,
358 |                                run_ackley: bool = True,
359 |                                run_rushhour: bool = True,
360 |                                tuned_cvrp_path=None,
361 |                                rushhour_hard_path=None) -> dict:
362 |     plan = load_final_plan(plan_path)
363 |     ok, errors = validate_final_plan(plan, require_official_data=True)
364 |     if not ok:
365 |         raise ValueError("final plan validation failed: " + " | ".join(errors))
366 |
367 |     tuned = None
368 |     if tuned_cvrp_path:
369 |         with open(tuned_cvrp_path) as f:
370 |             tuned = json.load(f)
371 |
372 |     output_dir = plan.get("output_dir", "results/final_experiments")
373 |     manifest = {
374 |         "plan_path": str(plan_path),
375 |         "resume": resume,
376 |         "run_cvrp": run_cvrp,
377 |         "run_ackley": run_ackley,
378 |         "run_rushhour": run_rushhour,
379 |         "tuned_cvrp_path": str(tuned_cvrp_path) if tuned_cvrp_path else "",
380 |         "tuned_cvrp_settings": tuned or {},
381 |         "alns_policy": (tuned or {}).get("alns_policy", {}),
382 |         "rushhour_hard_path": str(rushhour_hard_path) if rushhour_hard_path else "",
383 |         "cvrp_run_info": [],
384 |         "ackley_run_info": {},
385 |         "rushhour_run_info": {},
```

The submission audit output below is the real terminal output of the audit script on this repository — around 50 checks covering files, row counts, feasibility, report-vs-evidence consistency, and forbidden artifacts:

```
$ python scripts/audit_submission.py --check-results --check-pdf

PASS  file exists: README.md
PASS  file exists: requirements.txt
PASS  file exists: a3.py
PASS  file exists: run_a3.sh
PASS  file exists: run_a3.bat
PASS  file exists: report/assignment3_report.md
PASS  file exists: configs/final_experiment_plan.json
PASS  file exists: data/cvrp_bks.csv
PASS  file exists: report/evidence/cvrp_all_summary.csv
PASS  file exists: report/evidence/cvrp_all_policy_effective_summary.csv
PASS  file exists: report/evidence/ackley_d10_summary.csv
PASS  file exists: report/evidence/gp_gep_comparison_runs.csv
PASS  file exists: report/evidence/rushhour_hard_manual_summary.csv
PASS  file exists: report/evidence/rushhour_hard_gp_gep_summary.csv
PASS  file exists: report/evidence/final_execution_manifest.json
PASS  file exists: report/evidence/final_v3_summary.txt
PASS  file exists: report/evidence/cvrp_seed_robustness_summary.csv
PASS  file exists: report/evidence/direct_gp_gep_runs.csv
PASS  file exists: report/evidence/direct_manual_baselines.csv
PASS  file exists: report/evidence/direct_planner_manifest.json
PASS  file exists: report/evidence/cvrp_ils_runs.csv
PASS  file exists: report/evidence/cvrp_ils_summary.csv
PASS  file exists: report/evidence/cvrp_ils_manifest.json
PASS  file exists: report/assignment3_report.pdf
PASS  report has no unresolved placeholders
PASS  report has no overclaim phrases
PASS  report has an AI tools section
PASS  report has an explicit ILS section
PASS  report uses exploration/exploitation vocabulary
PASS  report has a references section
PASS  report has a cover identification block
PASS  report covers the tuned rerun
PASS  report covers the hard Rush Hour benchmark
PASS  report shows before/after tuning figure
PASS  report covers seed robustness
PASS  direct bonus manifest is not smoke
PASS  report covers the direct no-A* bonus
PASS  report shows the direct bonus figure
PASS  report has Stage 11 best gap 23.0063
PASS  report has Stage 11 best gap 2.9466
PASS  report has Stage 11 best gap 5.5139
PASS  report explains advanced local-search
PASS  report discloses the ALNS regression on X
PASS  report shows the advanced-impact figure
PASS  report B&B mean gap matches evidence (7.77%)
PASS  report B&B beats-baseline count matches evidence (evidence says 2/6)
PASS  B&B evidence CSVs agree with each other
PASS  report best gap for P-n16-k8 matches evidence (0.2967%)
PASS  report best gap for E-n22-k4 matches evidence (0.0746%)
PASS  report best gap for A-n32-k5 matches evidence (0.3931%)
PASS  report best gap for A-n80-k10 matches evidence (2.9466%)
PASS  report best gap for X-n101-k25 matches evidence (23.0063%)
PASS  report best gap for M-n200-k17 matches evidence (5.5139%)
PASS  result exists: results/final_experiments/raw/cvrp_all_instances.csv
PASS  result exists: results/final_experiments/summary/cvrp_all_summary.csv
PASS  result exists: results/final_experiments/raw/ackley_d10.csv
PASS  result exists: results/final_experiments/summary/ackley_d10_summary.csv
PASS  result exists: results/final_experiments/raw/gp_gep_comparison_runs.csv
PASS  result exists: results/final_experiments/raw/gp_gep_comparison_summary.txt
PASS  result exists: results/final_experiments/final_execution_manifest.json
PASS  cvrp_all_instances has 144 rows (144 rows)
PASS  all cvrp rows feasible
PASS  no cvrp row has errors
PASS  ackley_d10 has 21 rows (21 rows)
PASS  gp_gep_comparison has 6 rows (6 rows)
PASS  pdf size > 10 KB (1541160 bytes)
PASS  README documents the executable run commands
PASS  no dist/build binaries tracked
PASS  report matches evidence CSVs (verify_report_matches_csv) (report-vs-evidence: PASS (0 failed check(s)))
PASS  forbidden not tracked: docs
NOTE  docs exists locally but is untracked (will not be part of a git-based package)
PASS  forbidden not tracked: AI_USAGE.md
PASS  forbidden not tracked: report.md
PASS  forbidden not tracked: report.pdf
PASS  no ZIP files
PASS  removed instance name absent
NOTE  official .vrp files present: 6 (user-provided, not required to be tracked)

audit: PASS (0 failed check(s))
```

And the extracted report numbers, printed straight from the final CSVs — the same CVRP and Ackley numbers used in the tables above. (Two reading notes: the per-algorithm mean gaps in this printout average *all* raw rows, including both ALNS variants, so they differ by design from the policy-view table in Section 4; and its GP/GEP block prints the historical 4-puzzle smoke comparison, not the Section 8.1 hard benchmark.)

```
$ python scripts/extract_report_numbers.py

cvrp rows: 144, feasible: 144, with errors: 0
  best P-n16-k8: cvrp_sa cost 451.3351 gap 0.2967%
  best E-n22-k4: cvrp_sa cost 375.2798 gap 0.0746%
  best A-n32-k5: cvrp_alns cost 787.0819 gap 0.3931%
  best A-n80-k10: cvrp_alns_enhanced cost 1814.9490 gap 2.9466%
  best X-n101-k25: cvrp_ga_island cost 33938.6609 gap 23.0063%
  best M-n200-k17: cvrp_alns cost 1345.3021 gap 5.5139%
  mean gap baseline: 8.70%
  mean gap cvrp_aco: 7.74%
  mean gap cvrp_alns: 7.15%
  mean gap cvrp_alns_enhanced: 7.19%
  mean gap cvrp_bnb_lds: 7.77%
  mean gap cvrp_ga_island: 7.10%
  mean gap cvrp_sa: 7.43%
  mean gap cvrp_tabu: 8.12%
ackley rows: 21, errors: 0
  best ackley: ackley_alns seed 42 value 0.000000
gp/gep rows: 6
  best: gp_rushhour seed 42 eval_fitness 39956.0 solved 4
  expression diversity over all runs: 1.00
  summary txt:
    gp runs: 3
    gep runs: 3
    gp expression diversity: 1.00
    gep expression diversity: 1.00
    gep genome diversity: 1.00
    best gp expression: min((min(blocking, blocking) * (distance - 0)), ((distance * blocking) / (blocking / distance)))
    best gep expression: (5 * blocking)
    best gep genome: * 5.0 blocking min blocking log 0.5 0.0 5.0 1.0 distance 2.0 distance
    These results depend on the small puzzle set and the selected random seeds.
```

14. Conclusion

The implementation covers everything the assignment asks for: the six required search algorithms on both the Ackley warm-up and the six official CVRP instances, an explicit multi-stage CVRP heuristic with an honest feasibility-repair story, an explicit Iterated Local Search driver on top of them, and separate GP and GEP frameworks for evolving Rush Hour heuristics evaluated through A*. The six-algorithm final rerun (144 CVRP rows, all feasible), which combines the tuned settings with the Stage 11 advanced local-search moves, reached 0.07–0.39% on the three small instances, 2.95% on A-n80-k10, 5.51% on M-n200-k17, and 23.01% on the capacity-tight X-n101-k25 — still large and reported as a real limitation rather than smoothed over, along with the one algorithm-level regression the advanced pass caused (enhanced ALNS on X). The explicit ILS driver then pushed the overall project bests further on two instances (A-n80-k10 1.5219%, M-n200-k17 4.9072%, all 18 ILS runs feasible under the same budgets) and holds the lowest mean best gap (5.463%), while honestly failing to explore on X-n101-k25, where the tight packing rejects its kicks.

ALNS with the pre-declared hybrid policy keeps the lowest mean gap and beats the multi-stage baseline on all six instances, now joined by the memetic GA-Island; B&B/LDS, after gaining a small-instance deep-search mode, ties the best known project results on the two smallest instances but still returns its incumbent on the four larger ones, which is stated as-is. On Ackley (unchanged), the adapted ALNS

and B&B/LDS reached the optimum while untuned SA did not beat random search — a lesson the CVRP tuning then confirmed from the other direction. On the hard Rush Hour benchmark, both evolved frameworks beat the strongest manual heuristic by about 30% fewer A* expansions, while GP and GEP themselves remain honestly tied at the top with seed variance dominating. The optional no-A* bonus adds one more genuine result: GP evolved as a direct greedy planner solves 11/14 hard puzzles — beating the best hand-written direct policy (9/14) — while remaining honestly weaker than the A*-guided search (13/14). The main open improvements are full ejection chains for the capacity-tight CVRP instances (the Stage 11 swap/Or-opt/2-opt* moves recovered only part of the X-n101-k25 gap, and a packing-aware ejection-chain kick would also unfreeze the ILS perturbation there), a stronger bound to extend B&B/LDS beyond 25 customers, and pushing the hard Rush Hour set until the frameworks separate.

15. References

1. S. Bushinsky. *Artificial Intelligence Lab (203.3630), Experiment 3: Multi-Step Heuristics, Meta-Heuristics, Swarm Intelligence — CVRP.* Course assignment handout and lecture materials, University of Haifa, 2026.
2. G. Clarke and J. W. Wright. "Scheduling of Vehicles from a Central Depot to a Number of Delivery Points." *Operations Research* 12(4), 1964.
3. S. Kirkpatrick, C. D. Gelatt and M. P. Vecchi. "Optimization by Simulated Annealing." *Science* 220(4598), 1983.
4. F. Glover. "Tabu Search — Part I." *ORSA Journal on Computing* 1(3), 1989.
5. M. Dorigo and T. Stützle. *Ant Colony Optimization.* MIT Press, 2004.
6. S. Ropke and D. Pisinger. "An Adaptive Large Neighborhood Search Heuristic for the Pickup and Delivery Problem with Time Windows." *Transportation Science* 40(4), 2006.
7. W. D. Harvey and M. L. Ginsberg. "Limited Discrepancy Search." *Proceedings of IJCAI-95*, 1995.
8. H. R. Lourenço, O. C. Martin and T. Stützle. "Iterated Local Search." In *Handbook of Metaheuristics*, Kluwer, 2003.
9. D. H. Ackley. *A Connectionist Machine for Genetic Hillclimbing.* Kluwer, 1987 (the Ackley test function).
10. J. R. Koza. *Genetic Programming: On the Programming of Computers by Means of Natural Selection.* MIT Press, 1992.
11. C. Ferreira. "Gene Expression Programming: A New Adaptive Algorithm for Solving Problems." *Complex Systems* 13(2), 2001.
12. P. E. Hart, N. J. Nilsson and B. Raphael. "A Formal Basis for the Heuristic Determination of Minimum Cost Paths." *IEEE Transactions on Systems Science and Cybernetics* 4(2), 1968 (A*).
13. E. Uchoa, D. Pecin, A. Pessoa, M. Poggi, T. Vidal and A. Subramanian. "New Benchmark Instances for the Capacitated Vehicle Routing Problem." *European Journal of Operational Research* 257(3), 2017 — and the CVRPLIB collection, http://vrp.atd-lab.inf.puc-rio.br/ (source of the six official instances and BKS values).